

Bachelor's thesis

Hidden Markov Models for High Frequency FX Pairs Trading

Regime-gated co-integration strategies on currency tick data

Egil Furnes

egilsf@uio.no

Student number: 693784

Supervisor: Gudmund Horn Hermansen

STK-MAT2011 Project Work in Finance, Insurance, Risk and Data Analysis

Department of Mathematics

Faculty of Mathematics and Natural Sciences

15th May 2026

Abstract

This thesis asks whether Hidden Markov Models can improve a textbook pairs-trading rule on high-frequency FX tick data. The baseline is a standard z -score rule on a rolling cointegration spread between two related currency pairs. The regime model is a two-state Markov-switching AR(1) fitted to the same spread. We do not use the HMM as a separate forecaster. We use it only as a gate that switches the baseline off when the regime looks unfavourable. The trading direction is therefore the same as the baseline, and the only thing the HMM changes is when the rule is allowed to trade.

We compare four strategies on the same data: passive Buy & Hold, the baseline, an AR-gated variant, and a continuous MS-AR threshold. We run them on three currency-pair constructions across three one-month episodes, with a realistic cost model that charges the half-spread plus a small commission on every flip.

The baseline loses money on every pair-month: it trades too often, and the small per-trade alpha does not cover the cost. The HMM gate cuts trading activity and losses sharply, but the strategy never becomes profitable. The continuous MS-AR threshold does not beat the simpler binary gate. Passive Buy & Hold has the highest Sharpe ratio on most cells. The negative result is best read as a statement about the z -score rule under realistic FX costs: in this setting the HMM works as a defensive filter, not as a source of new return.

Acknowledgements

I would like to thank my supervisor, Associate Professor Gudmund Horn Hermansen (Department of Mathematics, University of Oslo), for his guidance and support throughout this project.

1 Introduction

Modern financial markets are largely mathematical objects. The trading pits and human judgement of the past have been replaced by electronic order books, statistical arbitrage engines, and high-speed execution algorithms (Aldridge 2013). Prices are no longer just summaries of economic fundamentals; they are dense data streams shaped by agents, algorithms, inventories, and news shocks (Hasbrouck 2007). The hard part is not speed but non-stationarity: any pattern that does exist is small, short-lived, and quickly traded away (Cont 2001). The practical question becomes whether a small statistical regularity can be found, tested, traded, and risk-managed after costs (López de Prado 2018).

Foreign exchange (FX) is a natural setting for this problem. It is global, electronic, and open almost twenty-four hours a day. On daily scales, exchange rates look like continuous diffusions (Black and Scholes 1973). At the tick level, prices move in small discrete jumps, quotes update asynchronously, and the path is dominated by microstructure frictions (Roll 1984). Figure 1 shows this scale dependence directly: the daily view is roughly diffusive, the hourly view shows local mean reversion, and the tick view is mostly bid-ask bounce. Currency pairs with shared economic exposure are natural candidates for pairs trading, where one bets on short-term deviations between two related exchange rates reverting to a common trend (Gatev et al. 2006). The three pairs we study (AUD/USD vs NZD/USD, EUR/NOK vs EUR/SEK, GBP/USD vs EUR/USD) span a range of liquidity and dependence.



Figure 1: The same FX series at three temporal resolutions. Top: daily closes, approximately diffusive. Middle: hourly bars, local mean reversion within a slow drift. Bottom: raw tick mid-prices, dominated by discrete bid-ask bounce and asynchronous quote updates.

A second difficulty is that each pair is itself non-stationary. Volatility clusters, correlations move around, and liquidity varies through the day (Cont 2001), so a rule that works in one period may fail in another. This is the language of market regimes: persistent hidden states, each with its own statistical behaviour, switching between one another by some random mechanism (Hamilton 1989). Because the regime is unobservable and we only see the data it produces, the natural tool is a Hidden Markov Model (HMM), where an observed series is driven by an unobserved Markov chain whose state determines the distribution of each observation (Rabiner 1989).

Role of the HMM in this thesis. The set-up is layered. We first build a single tradable signal: the rolling cointegration spread S_t (Engle and Granger 1987), standardised into a z -score Z_t , plugged into a textbook pairs-trading rule (Gatev et al. 2006). This baseline is back-tested under realistic transaction costs and serves as the benchmark that any more complex method has to beat. We then fit a Markov-switching AR(1) to the same S_t (Hamilton 1989) to recover a hidden chain K_t representing the active regime, and use it as a gate on top of the baseline rather than as a separate forecaster. The trading direction is always set by Z_t , and the entry rule is identical to the baseline; the HMM only decides whether the rule is allowed to act at a given bar. The regime-aware strategies cannot enter a trade the baseline does not enter, nor flip a position the baseline does not flip; they can only suppress trading in high-volatility regimes. Any difference in performance between the baseline and the regime-aware variants is therefore due to the gate alone, which is why baseline results are reported first. Figure 2 previews this regime recovery on a simulated two-regime AR(1).

Structure. Section 2 describes the data; Section 3 develops the cointegration spread and the baseline rule; Section 4 introduces the AR/HMM machinery and the regime-aware strategies; Section 5 specifies the walk-forward back-test and the performance metrics; Section 6 reports the empirical results; Section 7 concludes.



Figure 2: HMM regime recovery on a simulated two-regime AR(1). Top: observable series with the true generating regime shaded. Middle: smoothed posterior probability of the high-volatility regime from the Baum–Welch fit (Baum et al. 1970). Bottom: classification error. The hidden chain is recovered from the observable series alone, the same property used by the regime gate of Section 4.2 on the real cointegration spread.

2 Data and pre-processing

We use high-frequency bid–ask FX tick data from Dukascopy Bank SA (2026). The full pipeline (data processing, back-test driver, and all figures in this thesis) is implemented in Python and released as the open-source repository `stk-mat2011` in GitHub (Furnes 2026), implementation details, package versions, and reproduction scripts are documented there. The two legs of a pair are indexed by $i \in \{A, B\}$. The three pair constructions are AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, and GBP/USD–EUR/USD, chosen to span a range of liquidity, transaction costs, and pair-dependence (Krauss 2017).

Rather than running one continuous back-test, we study three independent one-month sub-samples: August 2024, December 2024, and April 2025. These months were chosen because all three pairs are actively moving in them, and because they cover very different macro backdrops: summer dollar weakness, year-end positioning, and the spring 2025 tariff episode. Each trading day inside a month is its own self-contained back-test. Every model and threshold introduced later (the rolling cointegration in Section 3, the z -score in Section 3, and the regime model in Section 4.1) is re-fitted from scratch at the start of every session, so any in-sample look-ahead is bounded by one day.

From bid and ask to mid and cost. Bid and ask streams are sorted by timestamp and aligned by a backward as-of merge. The bid and ask are used in two roles, never both at once. The *mid-price* (the average of bid and ask) is the input to every model in the rest of the thesis; it is the price the strategy sees when it decides whether to trade. The *half-spread* (half the bid–ask gap, in basis points) is not fed to any model; it is used in the back-test of Section 5 as the per-leg cost paid whenever the strategy crosses the

market. Writing $P_t^{\text{bid},i}$ and $P_t^{\text{ask},i}$ for the prevailing bid and ask of leg i at time t ,

$$\begin{aligned} P_t^{\text{mid},i} &= \frac{1}{2}(P_t^{\text{bid},i} + P_t^{\text{ask},i}), \\ \text{HS}_t^i &= \frac{1}{2} \frac{P_t^{\text{ask},i} - P_t^{\text{bid},i}}{P_t^{\text{mid},i}} \times 10^4. \end{aligned} \quad (1)$$

HS_t^i is the cost of crossing the spread on one leg (Roll 1984). A pairs trade enters and exits both legs at the same time, so the combined per-flip cost is $\text{HS}_t^A + \hat{\beta} \text{HS}_t^B$; whether the rolling rule of Section 3 can clear that cost is the central question of the empirical part of the thesis.

Pre-averaging on number of ticks. The raw mid-price is noisy on a tick-by-tick basis: bid–ask bounce, discreteness, and asynchronous quote updates all push the observed price around even when the underlying “efficient” price is unchanged (Hasbrouck 2007). We reduce this noise by pre-averaging in tick time, following Jacod et al. (2017): we group consecutive ticks into blocks of L ticks each and replace every block by the average mid-price inside it. If $P_\tau^{\text{mid},i}$ is the raw tick-level mid-price of leg i at tick τ , the pre-averaged mid-price at block index t is

$$\bar{P}_t^i = \frac{1}{L} \sum_{\tau=(t-1)L+1}^{tL} P_\tau^{\text{mid},i}, \quad t = 1, 2, \dots \quad (2)$$

The same block size L is used for both legs, so the pre-averaged legs stay synchronised. Pre-averaging in tick time is preferred to fixed-clock sampling because it gives an observation index that is more even in microstructure activity and lower in noise (Jacod et al. 2017). From here on, every quantity built from the mid-price (log-returns, cointegration spread, z -score, HMM observations) is built from $\bar{P}_t^i$, with t indexing blocks rather than ticks. To keep the notation light, we write $P_t^{\text{mid},i} \equiv \bar{P}_t^i$ in Sections 3–6.

3 Pairs trading with the baseline

This section defines the baseline strategy as a self-contained pairs-trading rule (Gatev et al. 2006). The regime-aware extensions of Section 4 inherit its entry direction and only change when it is allowed to trade.

Cointegration spread. Two non-stationary log-price series $\log P_t^{\text{mid},A}$ and $\log P_t^{\text{mid},B}$ are cointegrated if some linear combination of them is stationary (Engle and Granger 1987):

$$S_t = \log P_t^{\text{mid},A} - \beta \log P_t^{\text{mid},B}. \quad (3)$$

The hedge ratio β is fitted by the Engle–Granger two-step method: an OLS regression of $\log P^{\text{mid},A}$ on $\log P^{\text{mid},B}$, with the residuals tested for stationarity using an Augmented Dickey–Fuller test (Dickey and Fuller 1979). High-frequency FX is not globally cointegrated: β drifts with liquidity, time of day, and pair-specific shocks. We therefore estimate a rolling β_t over a window of W_β bars. Writing $x_s = \log P_s^{\text{mid},B}$, $y_s = \log P_s^{\text{mid},A}$, and $\mathcal{W}_t = \{t - W_\beta + 1, \dots, t\}$ with window means $\bar{x}_t, \bar{y}_t$,

$$\beta_t = \frac{\sum_{s \in \mathcal{W}_t} (x_s - \bar{x}_t)(y_s - \bar{y}_t)}{\sum_{s \in \mathcal{W}_t} (x_s - \bar{x}_t)^2}, \quad \alpha_t = \bar{y}_t - \beta_t \bar{x}_t. \quad (4)$$

The cointegration spread and its bar-by-bar return are

$$\begin{aligned} S_t &= \log P_t^{\text{mid}, A} - \beta_t \log P_t^{\text{mid}, B} - \alpha_t, \\ \Delta S_t &= r_t^A - \beta_{t-1} r_t^B, \end{aligned} \tag{5}$$

with bar log-returns $r_t^i = \log P_t^{\text{mid}, i} - \log P_{t-1}^{\text{mid}, i}$. The lag β_{t-1} in the return makes sure the hedge ratio used to weight the legs is already known when the return is realised.

Rolling z -score and the baseline rule. The baseline reads S_t through a rolling z -score over a window of length W_z ,

$$Z_t = \frac{S_t - \mu_t^S}{\sigma_t^S}, \tag{6}$$

with rolling mean and variance

$$\mu_t^S = \frac{1}{W_z} \sum_{s=t-W_z+1}^t S_s, \quad (\sigma_t^S)^2 = \frac{1}{W_z} \sum_{s=t-W_z+1}^t (S_s - \mu_t^S)^2.$$

With entry threshold z_q and exit threshold z_x , the position $\pi_t \in \{-1, 0, +1\}$ updates by

$$\pi_t = \begin{cases} +1 & \pi_{t-1} = 0, Z_t < -z_q, \\ -1 & \pi_{t-1} = 0, Z_t > +z_q, \\ 0 & \pi_{t-1} \neq 0, |Z_t| \leq z_x, \\ \pi_{t-1} & \text{otherwise.} \end{cases} \tag{7}$$

A long-spread position is opened when S_t is far enough below its rolling mean and closed once it comes back; the short rule is the mirror image. For every unit of leg A we trade $\hat{\beta}$ units of leg B , so the portfolio is exposed to the relative move between the two legs rather than to FX direction (Vidyamurthy 2004). Figure 3 shows the mechanics of a single round-trip on a representative pair.

As a benchmark we also track a do-nothing reference that simply holds a long unit spread position throughout the sample, $\pi_t^{\text{BH}} = +1$ for every t (Buy & Hold). It is the simplest possible answer to “what would happen if we did not trade at all?” It earns the bar-by-bar move ΔS_t of the cointegration spread, pays no transaction costs, and isolates the directional drift of the spread itself from any active timing. Any active rule that cannot beat this passive line on a risk-adjusted basis is, on the data we look at, not earning its costs.

The baseline assumes that S_t is generated by a single stationary process, so that one fixed pair of thresholds (z_q, z_x) captures “far from equilibrium” at every t . Two well-known features of FX tick data make this assumption fragile: the variance of ΔS_t clusters in time (conditional heteroskedasticity, (Cont 2001)), and the spread can drift directionally over hundreds of bars during news-driven episodes (Hasbrouck 2007). The fixed-threshold rule cannot tell these episodes apart from ordinary mean reversion, which motivates the regime-aware extensions of Section 4.

4 Regime-aware models

We introduce the AR/HMM machinery on a generic series Y_t and then apply it to the spread S_t . An autoregressive model of order p predicts Y_t from its p most recent past

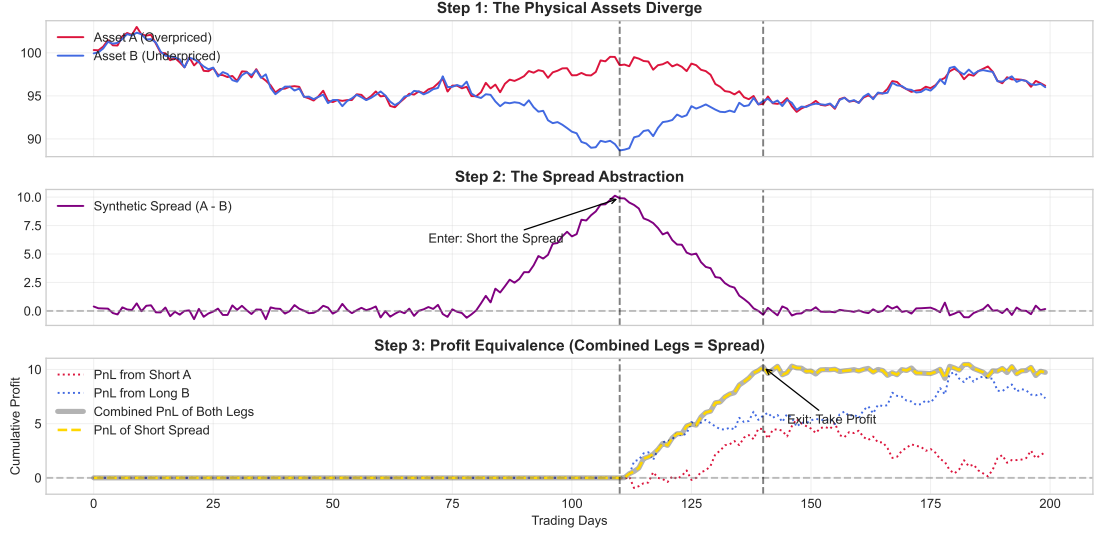


Figure 3: Mechanics of a single pairs-trade. Top: the price path of each leg over the trade window. Bottom: the combined long/short portfolio, which is exposed only to the relative move between the two legs and is approximately neutral to the common FX direction. The position is entered when the spread is far from its rolling mean (in z -score units) and closed as the spread reverts.

values (Hamilton 1994),

$$Y_t = c + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (8)$$

with the simplest non-trivial case AR(1),

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad (9)$$

which has unconditional mean $c/(1-\phi)$ when $|\phi| < 1$. AR(1) describes either momentum ($\phi > 0$) or fast mean reversion ($\phi < 0$), but only if the parameters (c, ϕ, σ^2) stay constant throughout the sample. We relax that constancy assumption by letting the parameters switch between a small set of values according to an unobserved Markov chain $K_t \in \{1, \dots, K\}$ (Hamilton 1989). The chain is described by a $K \times K$ transition matrix $\mathbf{P} = (p_{ij})$ and an initial distribution $\nu = (\nu_1, \dots, \nu_K)^\top$,

$$p_{ij} = \mathbb{P}(K_t = j \mid K_{t-1} = i), \quad \nu_k = \mathbb{P}(K_1 = k), \quad (10)$$

with row sums $\sum_{j=1}^K p_{ij} = 1$ for every i . The Markov assumption is deliberately simple: the future depends on the past only through the current state. But the model is still rich enough to capture persistence, sudden switching, and regime-dependent parameters in a single probabilistic story.

4.1 Markov-switching AR(1) on the spread

We now specialise to S_t from (5) and take $K = 2$. Conditional on the active regime $K_t = k$, the spread follows an AR(1) with its own intercept, AR coefficient, and innovation variance:

$$S_t \mid S_{t-1}, K_t = k \sim \mathcal{N}(c^{(k)} + \rho^{(k)} S_{t-1}, (\sigma^{(k)})^2),$$

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}. \quad (11)$$

All three regime-level parameters $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ switch with the chain (Krolzig 1997). For $|\rho^{(k)}| < 1$ the regime- k unconditional mean is $\mu^{(k)} = c^{(k)}/(1 - \rho^{(k)})$, which lets us write the same model in mean-deviation form,

$$S_t = \mu^{(k)} + \rho^{(k)}(S_{t-1} - \mu^{(k)}) + \varepsilon_t^{(k)}, \quad \varepsilon_t^{(k)} \sim \mathcal{N}(0, (\sigma^{(k)})^2). \quad (12)$$

A $\rho^{(k)}$ near zero means the spread reverts quickly; a $\rho^{(k)}$ near one means it behaves almost like a random walk. Before estimation, S_t is winsorised at $\text{median}(S) \pm 4 \cdot \text{std}(S)$ and rescaled by 10^4 for numerical stability.

Estimation, regime choice, and labelling. The HMM parameter set

$$\Theta = \{ \nu, \mathbf{P}, c^{(k)}, \rho^{(k)}, \sigma^{(k)} : k = 1, \dots, K \} \quad (13)$$

is fitted by maximum likelihood using the Markov-switching routine in the `statsmodels` package (Seabold and Perktold 2010), restarted from several random seeds to reduce sensitivity to local maxima. The same fit produces the smoothed regime posteriors

$$\gamma_t(k) = \mathbb{P}(K_t = k \mid S_1, \dots, S_T; \Theta). \quad (14)$$

We fix $K = 2$ throughout. A two-state model (one “normal” low-volatility regime and one “stress” high-volatility regime) is the standard parsimonious choice in the regime-switching literature (Hamilton 1989); in our preliminary exploration $K \in \{3, 4\}$ either failed to separate cleanly across seeds or split the high-volatility state into sub-regimes whose entry implications were indistinguishable at our cost level. With $K = 2$ we label the two regimes by innovation variance,

$$k_{\text{MR}} = \arg \min_k \sigma^{(k)}, \quad k_{\text{DR}} = \arg \max_k \sigma^{(k)}, \quad (15)$$

so the low-variance state is called the mean-reverting regime (MR) and the high-variance state the drifting regime (DR). We write $\gamma_t^{\text{MR}} = \mathbb{P}(K_t = k_{\text{MR}} \mid S_1, \dots, S_T)$ and $\gamma_t^{\text{DR}} = 1 - \gamma_t^{\text{MR}}$. In live trading only past data is available, so the back-test uses the filtered posterior $\mathbb{P}(K_t = k_{\text{MR}} \mid S_1, \dots, S_t)$ inside the gate; the smoothed posterior is kept for the diagnostic figures.

4.2 Regime-aware strategies

We build two regime-aware strategies on top of the baseline (7). Both keep the baseline’s entry direction and exit logic untouched: at every bar the candidate position is π_t^{baseline} from (7), and the regime model can only veto or attenuate that candidate. Neither variant can generate a position the baseline did not generate, nor reverse a position the baseline took.

AR-gated. A binary gate with danger threshold $\delta \in (0, 1)$,

$$G_t = \mathbb{I}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}, \quad (16)$$

modifies the position rule to

$$\pi_t = \begin{cases} 0 & \text{if } G_t = 0 \quad (\text{liquidate, stay flat}), \\ \pi_t^{\text{baseline}} & \text{if } G_t = 1. \end{cases} \quad (17)$$

The gate is binary: either trade exactly like the baseline, or sit in cash. It also closes existing positions: if we are long the spread and the HMM flags danger, we go flat at the next bar.

MS-AR. A continuous, regime-weighted entry threshold,

$$z_q^{\text{eff}}(t) = \gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v, \quad \text{with } z_v \geq z_q, \quad (18)$$

replaces z_q in (7) by $z_q^{\text{eff}}(t)$. When the MR regime is highly probable, $z_q^{\text{eff}} \approx z_q$ and the strategy trades like the baseline. When the DR regime dominates, the threshold rises towards z_v and the rule needs a bigger $|Z_t|$ deviation before it enters.

Table 1 summarises the four strategies compared throughout the rest of the thesis. The three active rules all share the same spread S_t and z -score Z_t , and differ only in whether and how γ_t^{MR} enters the decision. AR and MS-AR are not separate forecasters: on any bar where the HMM does not intervene they trade exactly like the baseline, so any difference between baseline and the regime-aware variants comes entirely from the gate suppressing or attenuating baseline trades.

Table 1: The four strategies compared in this thesis. The “Gate” column gives the indicator multiplying the baseline position π_t^{baseline} before it is taken; a value of 1 means the baseline is always allowed to trade.

Strategy	Always long?	Entry threshold	Gate	Reads HMM?
Buy & Hold	Yes	n/a	n/a	No
Baseline	No	z_q	1	No
AR	No	z_q	$\mathbb{1}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}$	Yes (binary)
MS-AR	No	$\gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v$	1	Yes (continuous)

5 Back-test protocol and metrics

Costs and walk-forward. Every position change in leg i pays the half-spread HS_t^i from (1) plus a small fixed commission (Aldridge 2013), so a round-trip on one pair-position costs roughly $\text{HS}^A + \hat{\beta} \text{HS}^B$ basis points. Strategies are evaluated walk-forward: the model parameters Θ and the hyperparameters $(W_\beta, W_z, z_q, z_x, z_v, \delta)$ are fitted on a training segment, applied to a disjoint out-of-sample test segment, and then the window is slid forward (López de Prado 2018).

Core performance metrics. We report three core metrics in the body: annualised volatility, the Sharpe ratio, and the maximum drawdown. A fuller set (skewness, excess kurtosis, VaR, CVaR, Sortino, Calmar, Ulcer index) is collected in Appendix 8.1 and reported in the per-pair tearsheets. Let R_t be the strategy return at bar t , $\mu_R = \mathbb{E}[R]$, $\sigma_R = \sqrt{\text{Var}(R)}$, V_t the portfolio value with running peak $V_t^* = \max_{s \leq t} V_s$, and R_f a benchmark risk-free rate (set to zero). The three core metrics are

$$\sigma_{\text{ann}} = \sigma_R \sqrt{T}, \quad \text{SR} = \frac{\mu_R - R_f}{\sigma_R}, \quad \text{MDD} = \max_t \frac{V_t^* - V_t}{V_t^*}, \quad (19)$$

covering scale (volatility), risk-adjusted return (Sharpe (Sharpe 1994)), and worst-case capital loss (maximum drawdown). Trade-level diagnostics (win rate, market exposure, profit factor, payoff ratio) summarise frequency and size; see Section 6 for their interpretation.

6 Empirical results and discussion

All numbers are net of the half-spread-plus-commission cost model of Section 5, with a fixed commission of 0.05 bps per flip and a per-day fit. For each of the three pair constructions we report three monthly episodes side by side (August 2024, December 2024, April 2025), so that cross-episode robustness stays visible. The full pair-month tearsheets, daily PnL ledgers, and per-pair figure-level diagnostics are in Appendix 8; the discussion below points to those tables rather than reproducing them.

6.1 Headline numbers

Table 2 reports total PnL in basis points for the four strategies across the nine pair-month cells. Table 3 reports trading activity and the annualised Sharpe ratio for the three active rules on the same cells. The full back-test grid (return, risk, risk-adjusted, trading, and distributional metrics) is in Tables 5–7 of Appendix 8.2, and the raw day-by-day ledgers are in Appendix 8.9 (Tables 10–12).

Table 2: Total PnL by pair, month, and strategy, in basis points, net of half-spread plus 0.05 bps commission. BH is Buy & Hold, BL the baseline cointegration rule, AR the binary AR-gated variant, and MS the continuous MS-AR threshold.

Pair	Month	BH	BL	AR	MS
AUD/USD–NZD/USD	Aug. 2024	–45.8	–966.5	–179.3	–206.1
	Dec. 2024	–39.3	–776.5	–136.7	–158.0
	Apr. 2025	+155.2	–2521.4	–652.2	–731.3
EUR/NOK–EUR/SEK	Aug. 2024	–148.5	–6277.4	–1816.7	–2490.5
	Dec. 2024	+60.6	–4473.8	–1592.8	–2070.8
	Apr. 2025	+527.7	–19906.4	–8103.5	–9487.3
GBP/USD–EUR/USD	Aug. 2024	+65.1	–396.1	–48.6	–60.1
	Dec. 2024	+25.1	–730.4	–69.6	–75.8
	Apr. 2025	+15.9	–1509.2	–350.6	–405.9

Table 3: Trading activity and Sharpe ratio for the three active strategies, by pair and month. “RT” is the number of round-trip trades; “Exp. %” is the percentage of bars in which the strategy holds a non-zero position; Sharpe is annualised (Sharpe 1994).

Pair	Month	Round-trips			Exposure (%)			Sharpe		
		BL	AR	MS	BL	AR	MS	BL	AR	MS
AUD/USD–NZD/USD	Aug. 2024	193	39	45	31	6	8	–33.8	–18.8	–14.3
	Dec. 2024	186	38	41	27	5	6	–36.9	–21.4	–18.2
	Apr. 2025	434	140	148	35	11	12	–58.6	–36.5	–36.4
EUR/NOK–EUR/SEK	Aug. 2024	606	137	158	35	7	9	–64.5	–45.7	–37.5
	Dec. 2024	435	158	181	35	12	15	–69.2	–58.4	–48.9
	Apr. 2025	1116	495	545	36	15	17	–126.1	–104.9	–104.5
GBP/USD–EUR/USD	Aug. 2024	281	79	83	34	7	9	–17.2	–5.9	–5.8
	Dec. 2024	325	58	63	35	6	7	–32.6	–12.3	–9.2
	Apr. 2025	590	154	161	36	9	10	–45.7	–32.6	–29.2

Three patterns show up in Tables 2–3 and repeat in every diagnostic below. First, the Baseline loses money on every pair-month. It generates between 186 and 1,116 round-trips, and its Sharpe ratios range from –17 on the cheapest cell to –126 on the

highest-spread one. The rule does earn a small per-trade alpha, but it pays the half-spread plus commission on every flip, and at this resolution the alpha is not enough to cover that cost (Krauss 2017). Second, the HMM gate cuts exposure by roughly four to six times and cumulative loss by about 75–85%, but it never turns the strategy profitable. The AR gate brings exposure down from 27–36% to 5–15%, and the MS-AR variant is one to three percentage points more permissive than AR on every cell. Third, passive Buy & Hold beats all three active rules on Sharpe in seven of the nine cells, and it is the only positive number in any column on EUR/NOK–EUR/SEK in December 2024 and April 2025 and on GBP/USD–EUR/USD in all three months.

Why does Buy & Hold dominate? The reason is mechanical, not statistical. Buy & Hold flips its position at most twice in a month (entry and exit), so it essentially never pays the half-spread plus commission of (1). The Baseline pays that cost on hundreds of flips per month. We also picked the three months precisely because all three pairs were actively moving, so the spread itself drifts visibly inside those months and a passive long-spread leg captures that drift for free. Buy & Hold therefore wins on Sharpe in cells where the active per-trade alpha is small enough to be eaten by the spread cost *and* the spread has enough directional drift in that month for a passive position to ride it; both conditions hold simultaneously on most of the nine cells in our sample.

6.2 Pair-level diagnostics and cointegration structure

The three pair constructions cover a wide range of liquidity (Roll 1984). GBP/USD–EUR/USD is the cheapest pair to trade, AUD/USD–NZD/USD sits in the middle, and EUR/NOK–EUR/SEK is the most expensive by a large margin. All six leg-level return series are strongly non-Gaussian, with high kurtosis and significant Ljung–Box statistics on squared returns, consistent with the volatility clustering and conditional heteroskedasticity commonly observed in financial returns (Cont 2001). The per-bar period-return panels in Appendix 8.6 show that these heavy tails remain visible at the trade-relevant horizon.

For a relative-value strategy, however, the central pre-trade question is not whether the individual legs are Gaussian. The relevant question is whether the constructed spread has enough statistical structure to justify mean-reversion trading. The cointegration screener therefore evaluates each pair-month along two dimensions. First, the full-sample test asks whether the entire monthly episode supports a stable long-run relation. Second, the rolling-window diagnostics ask whether this relation is persistent through time, or whether apparent cointegration is only a local and unstable feature of the sample.

The results in tables 4, 8 and 9 show that cointegration is episodic rather than universal. AUD/USD–NZD/USD has the cleanest full-sample evidence in December 2024, with a p-value of 0.0445 and a comparatively short full-sample half-life of 228.1 bars. EUR/NOK–EUR/SEK has the strongest rolling evidence in April 2025, where 23.1% of windows reject at the 5% level and 41.3% reject at the 10% level. GBP/USD–EUR/USD is generally the weakest cointegration candidate: its full-sample p-values remain high across all three months, and its rolling rejection rates are mostly close to zero. This is important because the later strategy results should not be interpreted as trades on permanently cointegrated assets. They are trades on spreads whose mean-reverting structure appears intermittently and must therefore be filtered, timed, and penalised heavily for transaction costs (Dickey and Fuller 1979; Newey and West 1987; Seabold and Perktold 2010; Krauss 2017).

A representative screener panel for AUD/USD–NZD/USD in August 2024 is shown in fig. 4. The complete August 2024 screener panels for all three pairs are reported in

Appendix 8.4.

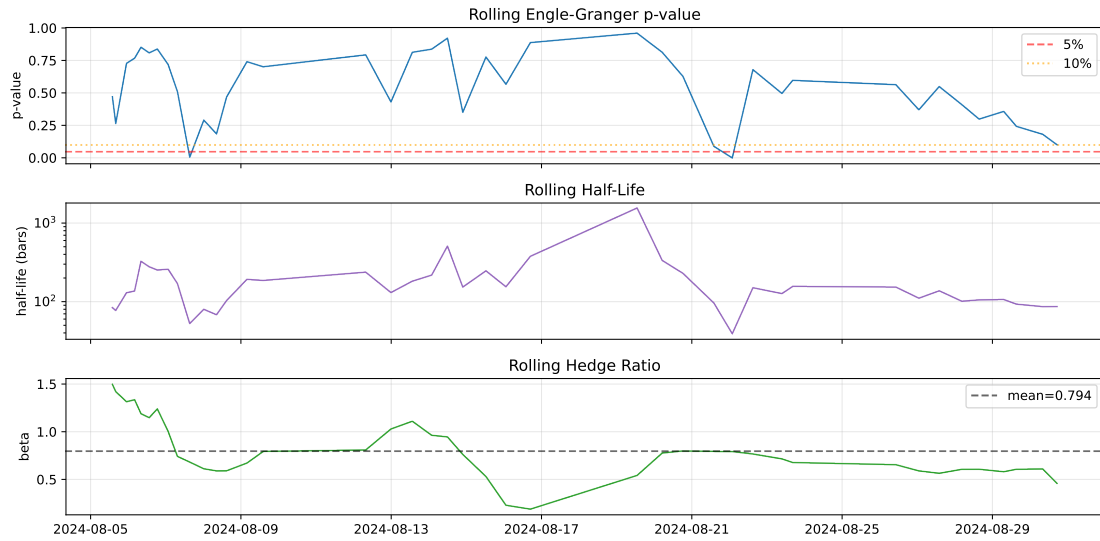


Figure 4: AUD/USD–NZD/USD, August 2024: cointegration screener diagnostics. The panel documents the estimated long-run relation between the two legs and the residual-spread behaviour used to assess whether the pair is statistically suitable for relative-value trading before strategy-level entry, exit, and regime-filtering rules are applied.

Table 4: Cointegration-screener diagnostics for AUD/USD–NZD/USD. The upper block reports full-sample diagnostics for each monthly episode. The lower block reports rolling-window diagnostics, where rejection shares measure the fraction of rolling windows with cointegration-test p-values below the stated threshold. HL denotes residual half-life in bars.

Full-sample diagnostics					
Month	<i>p</i> -value		HL	β	
Aug. 2024	0.5262		634.1	0.9092	
Dec. 2024	0.0445		228.1	0.8676	
Apr. 2025	0.1127		721.2	0.8588	
Rolling-window diagnostics					
Month	<i>p</i> < 0.05	<i>p</i> < 0.10	HL median	HL IQR	β mean $\pm$ sd
Aug. 2024	4.9%	7.3%	150.3	102–228	0.7942 $\pm$ 0.2970
Dec. 2024	0.0%	2.4%	121.6	96–144	0.9584 $\pm$ 0.1744
Apr. 2025	13.3%	26.7%	112.9	92–206	0.8908 $\pm$ 0.2999

The shape of the per-bar spread-return distribution makes the friction story concrete. Figure 5 shows, for AUD/USD–NZD/USD across the three episodes, the spread-return density on a linear y -axis (top row) and on a log y -axis (bottom row). The linear panels show a sharply peaked, near-zero-centred bulk, while the log panels expose Gaussian-incompatible tails at three or more standard deviations (Cont 2001). The corresponding grids for the other two pairs are in Appendix 8.7; on EUR/NOK–EUR/SEK the log-tail panels are visibly the widest, which previews the largest active-strategy losses we report below.

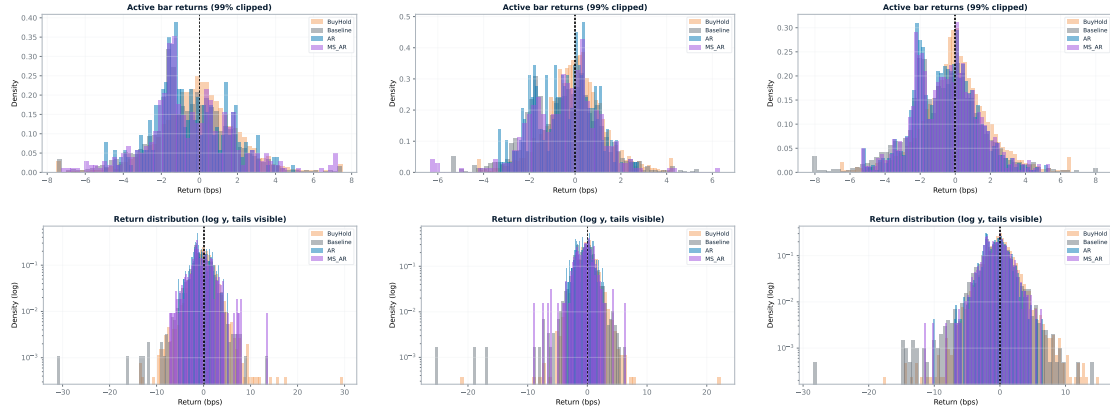


Figure 5: AUD/USD–NZD/USD spread-return distributions across the three episodes. Columns: August 2024 (left), December 2024 (centre), April 2025 (right). Top row: linear y -axis; bottom row: log y -axis exposing the Gaussian-incompatible tails of Cont (2001).

6.3 Baseline and HMM gate side by side

Figure 6 shows the cumulative profit-and-loss of the four strategies stacked across the three months for AUD/USD–NZD/USD. The corresponding panels for the other two pairs are collected on a single symlog grid in Appendix 8.3. Passive Buy & Hold drifts mildly, the Baseline (rolling z -score with $z_q = 2.0$, $z_x = 0$) bleeds steadily, and the two HMM-aware strategies sit in between, losing less than the Baseline but never crossing into profit.



Figure 6: AUD/USD–NZD/USD, three-month strategy comparison on a symmetric-log y -axis. Each panel stacks the four cumulative-PnL curves for one month under the half-spread plus 0.05bps commission cost model. The Baseline (blue) leaks steadily; the AR gate (green) and MS-AR threshold (red) trace a much flatter, less negative trajectory; Buy & Hold (grey) is positive in April 2025.

Three forces drive the Baseline result. The rolling z -score generates roughly 200 to 1,100 round-trips per month per pair, and each round-trip pays $HS_t^A + \hat{\beta} HS_t^B$ plus commission (Aldridge 2013). The per-trade alpha is small relative to that cost: the average winning trade is 1–1.5 bps on AUD/USD–NZD/USD and GBP/USD–EUR/USD and slightly larger on EUR/NOK–EUR/SEK, with comparable average losses (Appendix 8.2). The profit factor sits between 0.21 and 0.76 in every cell, the standard cost-of-mean-reversion-on-FX result of Krauss (2017) at tick resolution. And the loss distribution is heavily left-skewed and fat-tailed: Baseline skewness ranges from -0.29 to -13.0 and kurtosis from 12.5 to 308.6, so a handful of trade-level disasters dominates cumulative outcomes. April 2025 EUR/NOK–EUR/SEK is the most extreme case, with 1,116 round-trips, $-19,906$ bps realised, and a profit factor of 0.21.

The HMM gate inherits the Baseline’s entry direction and changes only when the rule is allowed to trade. The per-day Baum–Welch fits (Baum et al. 1970) separate the two regimes cleanly by innovation variance on every pair–day (Figures 15–17 in Appendix 8.8). The MR state has the smaller $\sigma^{(k)}$ and higher persistence p_{ii} ; the DR state has the larger $\sigma^{(k)}$ and smaller p_{ii} ; both means $\mu^{(k)}$ are close to zero, as the cointegration interpretation of S_t requires. This clean separation translates into a uniform drop in trading activity. The AR gate fires $G_t = 0$ in the drifting regime and shuts the Baseline off, pulling exposure from 27–36% down to 5–15%. The MS-AR threshold of (18) softens the binary switch into a continuous threshold interpolating between z_q and z_v , which is slightly more permissive: it raises exposure by one to three percentage points and round-trip count by five to ten. The same ordering shows up in risk: annualised volatility, max drawdown, and the Ulcer index of Martin and McCann (1989) all fall in the order $BL > MS > AR$ on every pair–month (Appendix 8.2). What the gate does *not* do is turn this lower exposure into a positive risk-adjusted return: the AR Sharpe is strictly less negative than the Baseline Sharpe everywhere, by a factor of two to five, but it never crosses zero. Once we condition on the rare drifting-regime entries the surviving sample is dominated by tail events, and the kurtosis of the gated strategies is actually larger than the Baseline’s. On this data, the HMM buys insurance, not alpha.

The continuous-threshold MS-AR adds one free hyperparameter, z_v , on top of the binary AR gate of (16), so that in-between values of γ_t^{MR} are used instead of being thrown away. The extra flexibility does not pay for itself. On every pair–month in Table 2, MS-AR loses slightly more than AR (for example -731.3 vs -652.2 bps on AUD/USD–NZD/USD April 2025): at $z_v = 3$ the continuous threshold admits a handful of marginal entries that the binary gate would have suppressed, and at our cost level those entries are net-negative. On two of nine cells MS-AR beats AR on Sharpe by a small margin, consistent with the warning of Cappé et al. (2005) that smoothing a regime decision rule is only useful when the marginal entries it admits are profitable in expectation. A BIC-style penalty (Schwarz 1978) on z_v , applied through a hyperparameter search such as Optuna (Akiba et al. 2019), would already select against MS-AR here.

6.4 Discussion

The four strategies tell a consistent story. Buy & Hold is the only strategy with positive cumulative PnL in a majority of pair–months, and the only one not paying a tick-level cost. Its positive outcomes (the largest is $+527.7$ bps on EUR/NOK–EUR/SEK April 2025, on the same data on which the Baseline loses $-19,906.4$ bps) reflect persistent directional drift in the spread rather than any traded signal (Gatev et al. 2006). The Baseline is the textbook cointegration rule and the worst-performing strategy on every

pair-month: the per-trade alpha is in the right direction (win rate is close to 40%, and several daily ledgers in Appendix 8 contain individual gross-positive days) but it does not clear the round-trip half-spread, reproducing the headline of Krauss (2017) on tick-level FX. AR delivers the gate’s promise (four-to-six-fold cut in exposure and round-trip count, 75–85% reduction in cumulative loss, proportionally smaller drawdowns and CVaRs); whether that counts as an improvement depends on the loss function, since on Sharpe it is still unambiguously bad. MS-AR substitutes a continuous threshold for the binary switch and is mildly worse on PnL but marginally better on Sharpe in two of nine cells, so the extra parameter z_v does not earn its keep at this cost level.

The qualitative pattern was partly expected: high-frequency FX is well known to be cost-dominated (Krauss 2017), and the conjecture from Section 1 that the HMM “decides when not to trade, not what to trade” lines up with the data. Less expected is the size of the Buy & Hold advantage in directional regimes like April 2025, and how cleanly the pair-level cost ordering shows up across all nine cells: GBP/EUR (cheapest, smallest loss), AUD/NZD (intermediate), EUR/NOK–SEK (most expensive, largest loss). That ordering is monotone in every month, and is the cleanest causal statement we can make from this data: the Baseline’s loss is, to leading order, the round-trip half-spread times the number of round-trips. The question posed in Section 1, whether the regime gate adds value over the baseline or only reduces exposure, therefore has a clear answer. The gate reduces exposure, but the Baseline does not carry enough net alpha at this cost level for that reduction to translate into a positive risk-adjusted return. The HMM is best read as a costed risk-reduction device that locates the point at which classical pairs-trading reasoning runs out of room on FX tick data. The natural next step is to pair the regime posteriors with an entry signal that is itself profitable after costs (a momentum signal in the spirit of Jegadeesh and Titman (1993), for example) rather than asking the HMM to rescue a baseline that is not.

7 Conclusion

This thesis tested whether Hidden Markov Models can add value to a classical pairs-trading rule on high-frequency FX tick data. The set-up is layered: the baseline is a textbook rolling z -score on a cointegration spread, and the HMM is used only as a regime gate that can switch the same baseline off. Four strategies, passive Buy & Hold, the baseline, an AR-gated variant, and a continuous MS-AR threshold, were compared on the same nine pair-month cells under a realistic cost model with per-day refits.

At a high level: the cointegration baseline is unprofitable on every pair-month at realistic costs; the HMM gate cuts exposure and cumulative loss sharply but never makes the strategy profitable; the continuous MS-AR variant does not improve on the simpler binary gate; and passive Buy & Hold beats the three active strategies on Sharpe in most cells. The regime model itself is well behaved, so the negative result is not driven by a misbehaving fit. It is driven by the cost-to-alpha ratio of the underlying z -score rule.

The exercise has clear limits. We use three pair constructions, three one-month episodes, $K = 2$ regimes, unit-size positions, a cost model that ignores latency and market impact, and an in-sample-smoothed gating posterior. The three episodes were also chosen for active price movement, which biases the comparison towards months where a passive long-spread position can drift directionally. The conclusion should therefore be read as a statement about the textbook z -score rule under realistic FX costs, not about HMMs as a class.

Three extensions follow naturally. First, pair the regime posteriors with an entry signal that is itself profitable after costs (a momentum signal in the drifting regime is the obvious candidate), rather than asking the HMM to rescue a mean-reversion signal that is already cost-dominated. Second, replace the per-day smoothed posterior with a strict walk-forward filter that uses only past observations; since this weakens the gate, the negative conclusion only tightens. Third, an HMM–GARCH hybrid that lets within-regime variance evolve smoothly is a natural fit for the fat tails seen in the spread returns. The full implementation, daily ledgers, and per-pair tearsheets are released as the open-source repository `stk-mat2011` as a starting point for follow-up work.

References

- Akiba, Takuya et al. (2019). ‘Optuna: A Next-generation Hyperparameter Optimization Framework’. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 2623–2631. DOI: [10.1145/3292500.3330701](https://doi.org/10.1145/3292500.3330701).
- Aldridge, Irene (2013). *High-Frequency Trading: A Practical Guide to Algorithmic Strategies and Trading Systems*. 2nd ed. Hoboken, NJ: Wiley.
- Artzner, Philippe et al. (1999). ‘Coherent Measures of Risk’. In: *Mathematical Finance* 9.3, pp. 203–228. DOI: [10.1111/1467-9965.00068](https://doi.org/10.1111/1467-9965.00068).
- Baum, Leonard E. et al. (1970). ‘A Maximization Technique Occurring in the Statistical Analysis of Probabilistic Functions of Markov Chains’. In: *The Annals of Mathematical Statistics* 41.1, pp. 164–171. DOI: [10.1214/aoms/1177697196](https://doi.org/10.1214/aoms/1177697196).
- Black, Fischer and Myron Scholes (1973). ‘The Pricing of Options and Corporate Liabilities’. In: *Journal of Political Economy* 81.3, pp. 637–654.
- Cappé, Olivier, Eric Moulines and Tobias Rydén (2005). *Inference in Hidden Markov Models*. Springer. DOI: [10.1007/b136369](https://doi.org/10.1007/b136369). URL: <https://doi.org/10.1007/b136369>.
- Cont, Rama (2001). ‘Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues’. In: *Quantitative Finance* 1.2, pp. 223–236.
- Dickey, David A. and Wayne A. Fuller (1979). ‘Distribution of the Estimators for Autoregressive Time Series with a Unit Root’. In: *Journal of the American Statistical Association* 74.366, pp. 427–431. DOI: [10.1080/01621459.1979.10482531](https://doi.org/10.1080/01621459.1979.10482531).
- Dukascopy Bank SA (2026). *Historical Data Feed*. Dukascopy Bank SA. URL: <https://www.dukascopy.com/swiss/english/marketwatch/historical/> (visited on 13/05/2026).
- Engle, Robert F. and Clive W. J. Granger (1987). ‘Co-integration and Error Correction: Representation, Estimation, and Testing’. In: *Econometrica* 55.2, pp. 251–276. DOI: [10.2307/1913236](https://doi.org/10.2307/1913236).
- Furnes, Egil (2026). *stk-mat2011: Code and reproducibility pipeline for “Hidden Markov Models for High Frequency FX Pairs Trading”*. URL: <https://github.com/egil10/stk-mat2011> (visited on 13/05/2026).
- Gatev, Evan, William N. Goetzmann and K. Geert Rouwenhorst (2006). ‘Pairs Trading: Performance of a Relative-Value Arbitrage Rule’. In: *Review of Financial Studies* 19.3, pp. 797–827. DOI: [10.1093/rfs/hhj020](https://doi.org/10.1093/rfs/hhj020).
- Hamilton, James D. (1989). ‘A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle’. In: *Econometrica* 57.2, pp. 357–384.
- (1994). *Time Series Analysis*. Princeton University Press. URL: <https://press.princeton.edu/books/hardcover/9780691042893/time-series-analysis>.
- Hasbrouck, Joel (2007). *Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading*. Oxford: Oxford University Press.
- Jacod, Jean et al. (2017). ‘Between Data Cleaning and Inference: Pre-averaging and Robust Estimators of the Efficient Price’. In: *Journal of Econometrics* 196.1, pp. 1–22. DOI: [10.1016/j.jeconom.2016.08.014](https://doi.org/10.1016/j.jeconom.2016.08.014). URL: <https://doi.org/10.1016/j.jeconom.2016.08.014>.
- Jegadeesh, Narasimhan and Sheridan Titman (1993). ‘Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency’. In: *Journal of Finance* 48.1, pp. 65–91. DOI: [10.1111/j.1540-6261.1993.tb04702.x](https://doi.org/10.1111/j.1540-6261.1993.tb04702.x).
- Krauss, Christopher (2017). ‘Statistical Arbitrage Pairs Trading Strategies: Review and Outlook’. In: *Journal of Economic Surveys* 31.2, pp. 513–545. DOI: [10.1111/joes.12153](https://doi.org/10.1111/joes.12153).

- Krolzig, Hans-Martin (1997). *Markov-Switching Vector Autoregressions: Modelling, Statistical Inference, and Application to Business Cycle Analysis*. Berlin: Springer. DOI: [10.1007/978-3-642-51684-9](https://doi.org/10.1007/978-3-642-51684-9).
- López de Prado, Marcos (2018). ‘Advances in Financial Machine Learning’. In: *Wiley*. Discusses information-driven bars including volume bars.
- Martin, Peter G. and Byron B. McCann (1989). ‘The Investor’s Guide to Fidelity Funds’. In: *New York: John Wiley & Sons*. Introduces the Ulcer index for drawdown analysis.
- Newey, Whitney K. and Kenneth D. West (1987). ‘A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix’. In: *Econometrica* 55.3, pp. 703–708. DOI: [10.2307/1913610](https://doi.org/10.2307/1913610).
- Rabiner, Lawrence R. (1989). ‘A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition’. In: *Proceedings of the IEEE* 77.2, pp. 257–286.
- Roll, Richard (1984). ‘A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market’. In: *Journal of Finance* 39.4, pp. 1127–1139. DOI: [10.1111/j.1540-6261.1984.tb03897.x](https://doi.org/10.1111/j.1540-6261.1984.tb03897.x).
- Schwarz, Gideon (1978). ‘Estimating the Dimension of a Model’. In: *The Annals of Statistics* 6.2, pp. 461–464. DOI: [10.1214/aos/1176344136](https://doi.org/10.1214/aos/1176344136).
- Seabold, Skipper and Josef Perktold (2010). ‘Statsmodels: Econometric and Statistical Modeling with Python’. In: *Proceedings of the 9th Python in Science Conference*, pp. 92–96. DOI: [10.25080/Majors-92bf1922-011](https://doi.org/10.25080/Majors-92bf1922-011).
- Sharpe, William F. (1994). ‘The Sharpe Ratio’. In: *Journal of Portfolio Management* 21.1, pp. 49–58. DOI: [10.3905/jpm.1994.409501](https://doi.org/10.3905/jpm.1994.409501).
- Sortino, Frank A. and Robert van der Meer (1991). ‘Downside Risk’. In: *Journal of Portfolio Management* 17.4, pp. 27–31. DOI: [10.3905/jpm.1991.409343](https://doi.org/10.3905/jpm.1991.409343).
- Vidyamurthy, Ganapathy (2004). *Pairs Trading: Quantitative Methods and Analysis*. Hoboken, NJ: Wiley.

8 Appendix

This appendix collects the quantitative tearsheets, figure-level diagnostics, and raw daily ledgers that go with the empirical results of Section 6. The body presents headline numbers and a small number of representative figures; the appendix is where the full back-test grid, the per-pair regime dynamics, and the trade-by-trade behaviour live. To keep the page count manageable we show each figure-level diagnostic for the August 2024 episode only, across all three pair constructions. The corresponding December 2024 and April 2025 figures, together with all raw outputs, daily tables, and plotting scripts, are in the open-source companion repository (Furnes 2026).

Reading the magnitudes. Several of the plots below show quantities that look numerically small at first glance, and we flag the units here so they are not misread as numerical artefacts. Per-bar spread returns ΔS_t are pre-averaged tick log-returns; on FX tick data they are naturally of order 10^{-5} to 10^{-4} per bar. That is the scale of high-frequency log-returns, not a sign of underflow or missing data. Cumulative PnL is reported in basis points ($\text{bps} = 10^{-4}$), and the symlog y -axis used in Section 8.3 compresses both very small and very large excursions onto the same panel by design. The pre-averaging step of Section 2 produces a regular grid of blocks, the cointegration estimation winsorises and rescales the spread for numerical stability (Section 4.1), and rows with missing bid or ask quotes are dropped before any model is fitted. The “small numbers” in the late plots therefore reflect the bar-level scale of FX tick data, not null values or floating-point underflow.

The appendix is organised in three blocks. The first block is the per-pair tearsheets in Section 8.2, one table per pair, stacking the three monthly episodes so that return, risk, risk-adjusted performance, trading activity, and distributional diagnostics can be read side by side for the four strategies on a single page. The second block is a sequence of figure grids, one plot type per page where possible: cross-pair cumulative-PnL panels on a symmetric-log axis in Section 8.3 (the same view of Figure 6 for the two pairs not shown in the body), rolling spread volatility in Section ??, per-bar period returns in Section 8.6, spread-return distributions for the two remaining pairs in Section 8.7, and per-day HMM regime dynamics in Section 8.8. The third block is the daily strategy summary tables of Section 8.9, one per pair, reporting every trading day in every episode at the round-trip level so that the bleed pattern of the Baseline and the gating behaviour of AR and MS can be inspected directly. The appendix closes with a symbol dictionary (Section 8.10) used as a lookup reference for the notation in the methods sections.

8.1 Additional performance metrics

This subsection defines the distributional, tail-risk, and extra risk-adjusted metrics that are reported in the per-pair tearsheets but kept out of Section 5 so the body could stay focused on the three core measures (volatility, Sharpe, max drawdown). Notation follows Section 5: R_t is the strategy return, μ_R and σ_R its mean and standard deviation, V_t the portfolio value with running peak V_t^* , and $R_f = 0$ the benchmark.

Distributional shape. Skewness and excess kurtosis describe the asymmetry and tail-heaviness of the return distribution:

$$\mathcal{S} = \frac{\mathbb{E}[(R - \mu_R)^3]}{\sigma_R^3}, \quad \mathcal{K} = \frac{\mathbb{E}[(R - \mu_R)^4]}{\sigma_R^4}. \quad (20)$$

Tail risk. With $\alpha = 0.95$, the Value-at-Risk, Conditional VaR, and tail-fatness ratio are (Artzner et al. 1999)

$$\text{VaR}_\alpha = -Q_{1-\alpha}(R), \quad \text{CVaR}_\alpha = -\mathbb{E}[R \mid R \leq -\text{VaR}_\alpha], \quad \text{TF} = \frac{\text{CVaR}_\alpha}{\text{VaR}_\alpha}. \quad (21)$$

Drawdown family beyond MDD. The Ulcer index of Martin and McCann (1989) measures the depth and duration of drawdowns by the root-mean-square of the drawdown series $D_t = (V_t^* - V_t)/V_t^*$:

$$\text{UI} = \sqrt{\frac{1}{N} \sum_{t=1}^N D_t^2}. \quad (22)$$

Additional risk-adjusted ratios. The Sortino ratio (Sortino and Meer 1991) divides by downside deviation σ_d rather than total volatility, and the Calmar ratio divides annualised return AR by the worst observed drawdown:

$$\text{Sortino} = \frac{\mu_R - R_f}{\sigma_d}, \quad \text{Calmar} = \frac{\text{AR}}{\text{MDD}}. \quad (23)$$

8.2 Tearsheets

The five blocks in the tables below read as follows. The Return block reports the realised cumulative PnL in basis points over the episode (Total Return) and its annualised counterpart (Annual Return, computed as $\text{bps} \times T_{\text{ann}}/T_{\text{episode}}$). The Risk block reports annualised volatility σ_{ann} of (19), the maximum drawdown MDD and its duration in bars, the Ulcer index UI of (22), and the 95% Value-at-Risk and Conditional VaR of (21). The Risk-adj. block reports the Sharpe ratio of (19) and the Sortino and Calmar ratios of (23), together with the trade-level Profit Factor (sum of winning trades divided by absolute sum of losing trades), Payoff Ratio (average win divided by average loss), and a Tail Ratio (the ratio of the 95th to the absolute 5th percentile of returns). The Trading block reports the number of round-trip trades, the win rate, the market exposure (fraction of bars with a non-zero position), and the average winning and losing trade in basis points. The Distr. block reports the per-bar skewness $\mathcal{S}$ and kurtosis $\mathcal{K}$ of (20), both of which are large in magnitude under the conditionally heteroskedastic, heavy-tailed return distribution discussed in Section 6.2.

Table 5: AUD/USD–NZD/USD strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS		
Return	Aug. 2024	Total Return (bps)	-45.84	-966.47	-179.31	-206.06		
		Annual Return (bps)	-577.34	-12172.56	-2258.32	-2595.27		
	Dec. 2024	Total Return (bps)	-39.34	-776.45	-136.70	-158.02		
		Annual Return (bps)	-495.49	-9779.28	-1721.69	-1990.25		
	Apr. 2025	Total Return (bps)	155.23	-2521.35	-652.23	-731.32		
		Annual Return (bps)	1955.13	-31756.03	-8214.73	-9210.85		
Risk	Aug. 2024	Annual Volatility (bps)	554.99	360.20	120.07	181.98		
		Max Drawdown (bps)	-171.64	-966.67	-179.31	-210.59		
		Max DD Duration (bars)	2302	5330	4741	4999		
		Ulcer Index	70.08	613.89	144.12	167.35		
		VaR 95% (bps)	-3.29	-2.37	0.00	0.00		
		CVaR 95% (bps)	-4.87	-4.08	-0.06	-0.10		
	Dec. 2024	Annual Volatility (bps)	356.59	264.79	80.61	109.32		
		Max Drawdown (bps)	-119.60	-778.78	-136.70	-162.72		
		Max DD Duration (bars)	2698	5435	5321	5313		
		Ulcer Index	49.36	412.81	68.77	80.60		
		VaR 95% (bps)	-2.09	-1.85	0.00	0.00		
		CVaR 95% (bps)	-3.09	-2.99	-0.04	-0.05		
	Apr. 2025	Annual Volatility (bps)	743.57	541.55	225.04	252.92		
		Max Drawdown (bps)	-198.20	-2521.66	-652.45	-737.07		
		Max DD Duration (bars)	3452	10764	9796	10469		
		Ulcer Index	64.99	1455.73	461.45	513.13		
		VaR 95% (bps)	-3.21	-2.68	-0.64	-0.96		
		CVaR 95% (bps)	-4.58	-4.44	-2.06	-2.32		
Risk-adj.	Aug. 2024	Sharpe Ratio	-1.04	-33.79	-18.81	-14.26		
		Sortino Ratio	-1.45	-24.89	-7.84	-7.03		
		Calmar Ratio	3.36	12.59	12.59	12.32		
		Profit Factor	0.99	0.51	0.47	0.59		
		Payoff Ratio	1.00	0.81	0.86	0.93		
		Tail Ratio	0.98	0.63	-	-		
	Dec. 2024	Sharpe Ratio	-1.39	-36.93	-21.36	-18.21		
		Sortino Ratio	-1.85	-23.03	-7.44	-5.80		
		Calmar Ratio	4.14	12.56	12.59	12.23		
		Profit Factor	0.98	0.44	0.38	0.44		
		Payoff Ratio	0.97	0.67	0.55	0.62		
		Tail Ratio	0.99	0.48	-	-		
	Apr. 2025	Sharpe Ratio	2.63	-58.64	-36.50	-36.42		
		Sortino Ratio	3.71	-46.67	-18.38	-19.30		
		Calmar Ratio	9.86	12.59	12.59	12.50		
		Profit Factor	1.02	0.46	0.44	0.47		
		Payoff Ratio	1.03	0.73	0.71	0.70		
		Tail Ratio	0.99	0.55	0.00	0.00		
Trading	Aug. 2024	Number of Trades	0	193	39	45		
		Win Rate	49.71%	38.82%	35.18%	38.95%		
		Market Exposure	89.78%	31.25%	5.70%	7.82%		
		Avg Win (bps)	1.60	1.56	1.45	1.83		
		Avg Loss (bps)	-1.60	-1.93	-1.69	-1.97		
		Number of Trades	0	186	38	41		
	Dec. 2024	Win Rate	50.35%	39.65%	40.78%	41.39%		
		Market Exposure	90.43%	27.03%	5.14%	6.03%		
		Avg Win (bps)	0.99	1.04	0.72	0.89		
		Avg Loss (bps)	-1.02	-1.55	-1.32	-1.45		
		Number of Trades	0	434	140	148		
		Win Rate	49.89%	38.96%	38.43%	39.86%		
	Apr. 2025	Market Exposure	94.92%	35.23%	10.50%	12.07%		
		Avg Win (bps)	1.49	1.47	1.18	1.23		
		Avg Loss (bps)	-1.45	-2.02	-1.67	-1.74		
		Distr.	Aug. 2024	Skewness	0.56	-3.15	-2.65	0.73
				Kurtosis	11.73	61.41	45.38	67.15
			Dec. 2024	Skewness	0.06	-5.94	-4.67	-5.81
Kurtosis	26.41			116.15	50.27	119.04		
Apr. 2025	Skewness		0.08	-2.06	-2.50	-2.45		
	Kurtosis		4.87	28.09	34.11	34.01		

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

Table 6: EUR/NOK–EUR/SEK strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS
Return	Aug. 2024	Total Return (bps)	-148.51	-6277.42	-1816.72	-2490.51
		Annual Return (bps)	-1870.45	-79062.97	-22881.30	-31367.54
	Dec. 2024	Total Return (bps)	60.58	-4473.83	-1592.84	-2070.78
		Annual Return (bps)	763.03	-56347.12	-20061.51	-26081.12
	Apr. 2025	Total Return (bps)	527.70	-19906.39	-8103.49	-9487.28
		Annual Return (bps)	6646.26	-250717.51	-102062.01	-119490.65
Risk	Aug. 2024	Annual Volatility (bps)	904.87	1226.06	501.14	836.51
		Max Drawdown (bps)	-459.55	-6277.42	-1816.72	-2492.83
		Max DD Duration (bars)	12611	14774	14236	14774
		Ulcer Index	316.62	3826.73	1139.70	1583.40
		VaR 95% (bps)	-3.27	-3.52	0.00	-0.12
		CVaR 95% (bps)	-4.99	-7.99	-0.14	-3.97
	Dec. 2024	Annual Volatility (bps)	634.41	813.82	343.78	532.97
		Max Drawdown (bps)	-212.66	-4473.83	-1592.84	-2070.78
		Max DD Duration (bars)	4166	10633	10500	10500
		Ulcer Index	72.69	2498.43	1004.40	1232.91
		VaR 95% (bps)	-2.58	-4.28	-0.39	-0.75
		CVaR 95% (bps)	-4.19	-6.97	-3.39	-4.50
	Apr. 2025	Annual Volatility (bps)	1508.79	1988.38	973.31	1143.00
		Max Drawdown (bps)	-474.38	-19906.39	-8103.49	-9487.28
		Max DD Duration (bars)	18798	26085	26085	26085
		Ulcer Index	294.93	11686.94	4533.58	5341.94
		VaR 95% (bps)	-3.56	-7.34	-0.68	-1.42
		CVaR 95% (bps)	-6.35	-11.93	-6.53	-7.71
Risk-adj.	Aug. 2024	Sharpe Ratio	-2.07	-64.49	-45.66	-37.50
		Sortino Ratio	-2.66	-35.21	-11.67	-9.99
		Calmar Ratio	4.07	12.59	12.59	12.58
		Profit Factor	0.99	0.32	0.11	0.16
		Payoff Ratio	0.98	0.49	0.18	0.24
		Tail Ratio	1.00	0.39	–	0.00
	Dec. 2024	Sharpe Ratio	1.20	-69.24	-58.36	-48.94
		Sortino Ratio	1.51	-40.33	-21.63	-17.54
		Calmar Ratio	3.59	12.59	12.59	12.59
		Profit Factor	1.01	0.27	0.14	0.18
		Payoff Ratio	1.03	0.42	0.21	0.26
		Tail Ratio	1.03	0.24	0.00	0.14
	Apr. 2025	Sharpe Ratio	4.41	-126.09	-104.86	-104.54
		Sortino Ratio	5.30	-76.34	-41.53	-44.25
		Calmar Ratio	14.01	12.59	12.59	12.59
		Profit Factor	1.03	0.21	0.08	0.11
		Payoff Ratio	1.03	0.35	0.14	0.19
		Tail Ratio	1.02	0.15	0.06	0.09
Trading	Aug. 2024	Number of Trades	0	606	137	158
		Win Rate	50.05%	39.65%	38.10%	39.15%
		Market Exposure	96.29%	34.84%	7.25%	9.22%
		Avg Win (bps)	1.38	1.45	0.54	0.86
		Avg Loss (bps)	-1.40	-2.97	-3.06	-3.55
	Dec. 2024	Number of Trades	0	435	158	181
		Win Rate	49.42%	39.48%	40.00%	40.44%
		Market Exposure	95.08%	35.19%	11.90%	14.64%
		Avg Win (bps)	1.13	1.14	0.50	0.71
		Avg Loss (bps)	-1.09	-2.71	-2.42	-2.70
	Apr. 2025	Number of Trades	0	1116	495	545
		Win Rate	50.08%	37.89%	36.63%	36.97%
		Market Exposure	97.90%	36.22%	14.85%	17.44%
		Avg Win (bps)	1.41	1.50	0.50	0.71
		Avg Loss (bps)	-1.38	-4.30	-3.59	-3.72
Distr.	Aug. 2024	Skewness	0.21	-13.00	-15.75	-22.55
		Kurtosis	21.37	308.63	415.35	740.30
	Dec. 2024	Skewness	-0.17	-10.48	-5.90	-15.88
		Kurtosis	12.07	251.63	40.05	458.00
	Apr. 2025	Skewness	0.48	-5.59	-6.44	-6.48
		Kurtosis	38.58	67.41	60.79	70.40

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

Table 7: GBP/USD–EUR/USD strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS
Return	Aug. 2024	Total Return (bps)	65.09	-396.06	-48.60	-60.14
		Annual Return (bps)	819.75	-4988.30	-612.05	-757.45
	Dec. 2024	Total Return (bps)	25.10	-730.39	-69.62	-75.81
		Annual Return (bps)	316.08	-9199.19	-876.90	-954.82
	Apr. 2025	Total Return (bps)	15.94	-1509.17	-350.61	-405.94
		Annual Return (bps)	200.78	-19007.67	-4415.92	-5112.73
Risk	Aug. 2024	Annual Volatility (bps)	497.41	289.35	103.71	130.86
		Max Drawdown (bps)	-171.91	-417.06	-67.12	-67.90
		Max DD Duration (bars)	5168	6985	6156	6985
		Ulcer Index	85.73	263.30	41.59	42.89
		VaR 95% (bps)	-2.56	-1.60	-0.00	-0.01
		CVaR 95% (bps)	-3.74	-2.77	-0.04	-1.08
	Dec. 2024	Annual Volatility (bps)	444.49	282.61	71.07	103.73
		Max Drawdown (bps)	-101.94	-738.06	-71.95	-79.82
		Max DD Duration (bars)	2879	8011	8011	8063
		Ulcer Index	51.72	401.64	47.07	55.06
		VaR 95% (bps)	-2.12	-1.48	0.00	0.00
		CVaR 95% (bps)	-3.22	-2.63	-0.03	-0.03
	Apr. 2025	Annual Volatility (bps)	644.21	416.02	135.39	174.93
		Max Drawdown (bps)	-332.36	-1525.73	-356.08	-420.51
		Max DD Duration (bars)	7736	14733	13115	14105
		Ulcer Index	146.04	906.32	212.03	249.97
		VaR 95% (bps)	-2.34	-1.65	-0.09	-0.27
		CVaR 95% (bps)	-3.36	-2.85	-0.94	-1.19
Risk-adj.	Aug. 2024	Sharpe Ratio	1.65	-17.24	-5.90	-5.79
		Sortino Ratio	2.26	-14.69	-2.52	-2.76
		Calmar Ratio	4.77	11.96	9.12	11.16
		Profit Factor	1.02	0.76	0.83	0.84
		Payoff Ratio	1.00	1.04	1.03	1.04
		Tail Ratio	0.99	0.85	–	0.00
	Dec. 2024	Sharpe Ratio	0.71	-32.55	-12.34	-9.20
		Sortino Ratio	0.95	-25.99	-4.99	-3.39
		Calmar Ratio	3.10	12.46	12.19	11.96
		Profit Factor	1.01	0.61	0.65	0.72
		Payoff Ratio	1.00	0.93	1.03	1.07
		Tail Ratio	1.02	0.71	–	–
	Apr. 2025	Sharpe Ratio	0.31	-45.69	-32.62	-29.23
		Sortino Ratio	0.45	-37.03	-14.91	-13.20
		Calmar Ratio	0.60	12.46	12.40	12.16
		Profit Factor	1.00	0.60	0.50	0.55
		Payoff Ratio	1.01	0.92	0.86	0.89
		Tail Ratio	1.01	0.68	0.00	0.00
Trading	Aug. 2024	Number of Trades	0	281	79	83
		Win Rate	50.48%	42.26%	44.49%	44.87%
		Market Exposure	92.27%	33.50%	7.39%	9.18%
		Avg Win (bps)	1.23	1.26	1.00	1.10
		Avg Loss (bps)	-1.24	-1.21	-0.96	-1.06
	Dec. 2024	Number of Trades	0	325	58	63
		Win Rate	50.28%	39.48%	38.66%	40.29%
		Market Exposure	93.54%	35.31%	5.86%	6.84%
		Avg Win (bps)	1.01	1.01	0.70	0.86
		Avg Loss (bps)	-1.01	-1.08	-0.68	-0.81
	Apr. 2025	Number of Trades	0	590	154	161
		Win Rate	49.78%	39.64%	36.73%	38.33%
		Market Exposure	96.29%	35.64%	8.60%	10.13%
		Avg Win (bps)	1.09	1.09	0.75	0.87
		Avg Loss (bps)	-1.08	-1.19	-0.87	-0.98
Distr.	Aug. 2024	Skewness	-0.01	-0.29	-0.49	0.18
		Kurtosis	11.41	12.48	52.25	54.06
	Dec. 2024	Skewness	-0.32	-1.16	0.90	-1.03
		Kurtosis	9.58	25.35	122.03	334.80
	Apr. 2025	Skewness	0.25	-1.15	-1.83	-2.87
		Kurtosis	4.75	20.62	37.78	66.55

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

8.3 Three-month cumulative-PnL panels (symlog)

Symlog-scaled cumulative-PnL panels stacking the four strategies (Buy & Hold, Baseline, AR, MS-AR) across the three monthly episodes for each of the three pair constructions.

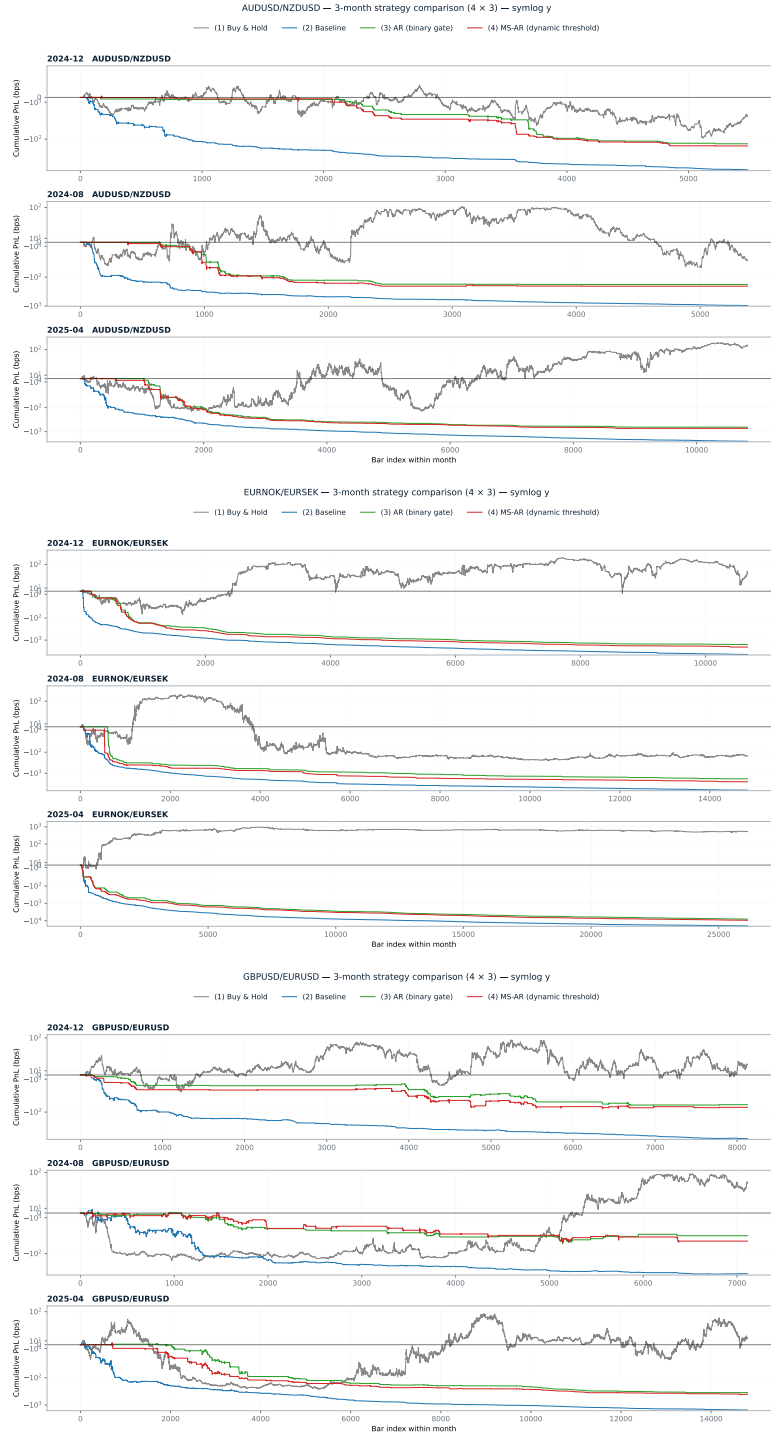


Figure 7: Three-month strategy comparison on a symmetric-log y -axis. Top: AUD/USD–NZD/USD; middle: EUR/NOK–EUR/SEK; bottom: GBP/USD–EUR/USD. Each panel stacks the four cumulative-PnL curves across the three monthly episodes (August 2024, December 2024, April 2025).

8.4 Cointegration screener diagnostics

This appendix reports the cointegration-screening diagnostics for the August 2024 episode of each currency-pair spread. The plots document the pre-trade statistical structure used by the spread engine: whether the two legs share a stable long-run relation, whether the residual spread is plausibly stationary and mean-reverting, and whether the pair is defensible as a relative-value trading candidate.

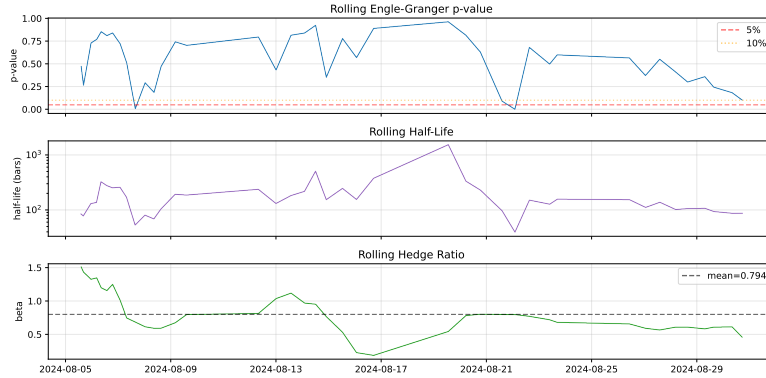


Figure 8: AUD/USD–NZD/USD, August 2024: cointegration screener diagnostics.

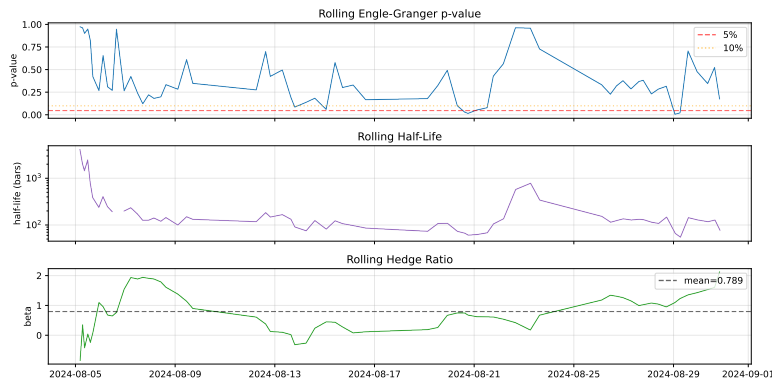


Figure 9: EUR/NOK–EUR/SEK, August 2024: cointegration screener diagnostics.

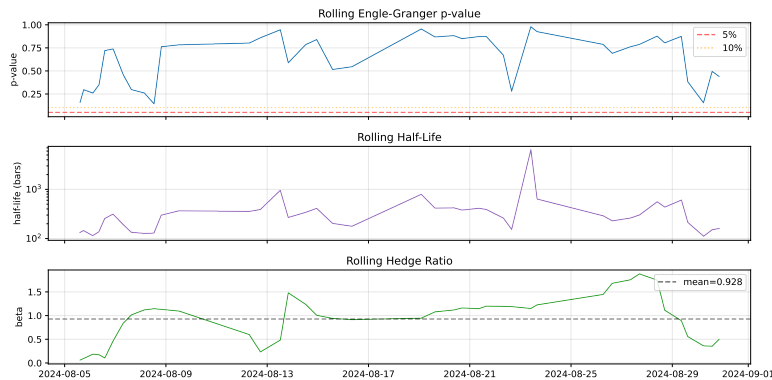


Figure 10: GBP/USD–EUR/USD, August 2024: cointegration screener diagnostics.

8.5 Cointegration tables

Table 8: Cointegration-screener diagnostics for EUR/NOK–EUR/SEK. The pair exhibits the strongest rolling evidence in April 2025, but it is also the most expensive pair to trade. Statistical mean reversion is therefore not sufficient on its own: the signal must survive the substantially higher transaction-cost burden. HL denotes residual half-life in bars.

Full-sample diagnostics					
Month	p-value		HL	β	
Aug. 2024	0.4452		1358.2	1.0603	
Dec. 2024	0.2909		690.6	−1.0217	
Apr. 2025	0.0170		670.2	1.5689	
Rolling-window diagnostics					
Month	$p < 0.05$	$p < 0.10$	HL median	HL IQR	β mean $\pm$ sd
Aug. 2024	6.2%	12.3%	129.2	108–175	0.7893 $\pm$ 0.6505
Dec. 2024	4.5%	6.8%	191.0	158–279	0.2394 $\pm$ 0.8960
Apr. 2025	23.1%	41.3%	100.0	49–149	0.8746 $\pm$ 0.5419

Table 9: Cointegration-screener diagnostics for GBP/USD–EUR/USD. Despite being the cheapest pair to trade, the spread has weak cointegration evidence across the selected monthly episodes. Full-sample *p*-values are high, and rolling-window rejection rates are limited, especially in August and December 2024. HL denotes residual half-life in bars.

Full-sample diagnostics					
Month	p-value		HL	β	
Aug. 2024	0.5879		1335.8	1.3084	
Dec. 2024	0.2558		599.1	1.0124	
Apr. 2025	0.7905		3845.5	0.7376	
Rolling-window diagnostics					
Month	$p < 0.05$	$p < 0.10$	HL median	HL IQR	β mean $\pm$ sd
Aug. 2024	0.0%	0.0%	287.7	160–407	0.9283 ± 0.4813
Dec. 2024	0.0%	7.3%	301.9	128–488	0.7682 ± 0.2791
Apr. 2025	9.2%	10.8%	201.0	131–335	0.6677 ± 0.4401

8.6 Period returns

Per-bar period returns for the August 2024 episode of each pair. The remaining months are available in the companion repository (Furnes [2026](#)).

8.6.1 AUD/USD–NZD/USD

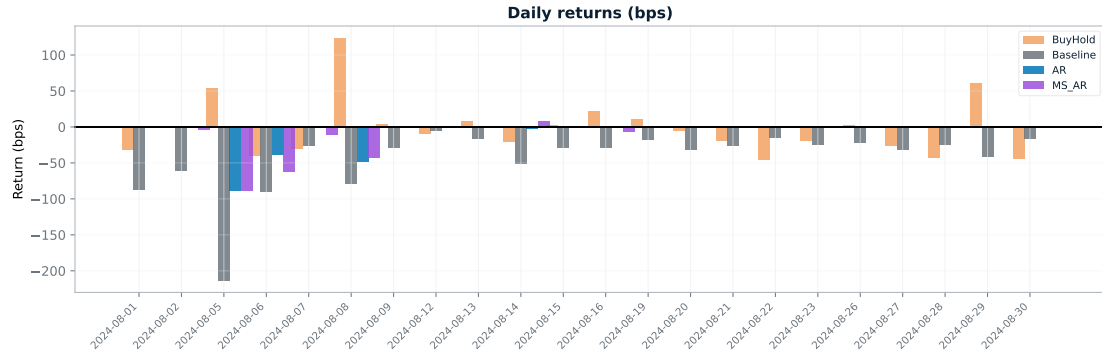


Figure 11: AUD/USD–NZD/USD, August 2024: period returns.

8.6.2 EUR/NOK–EUR/SEK

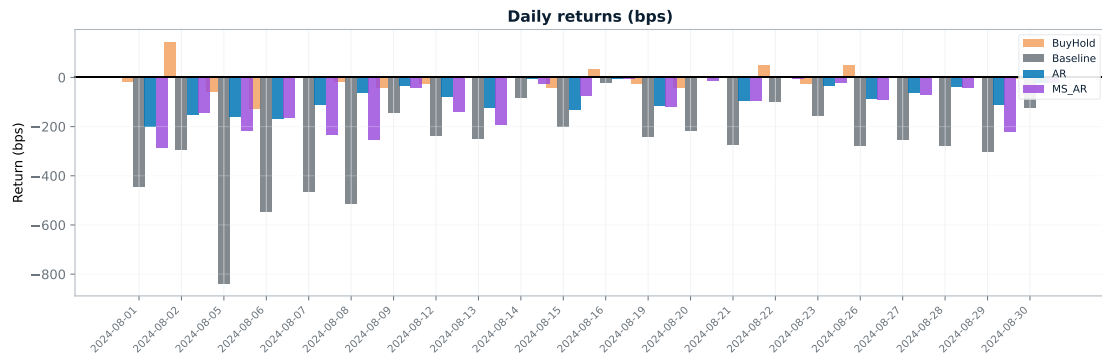


Figure 12: EUR/NOK–EUR/SEK, August 2024: period returns.

8.6.3 GBP/USD–EUR/USD

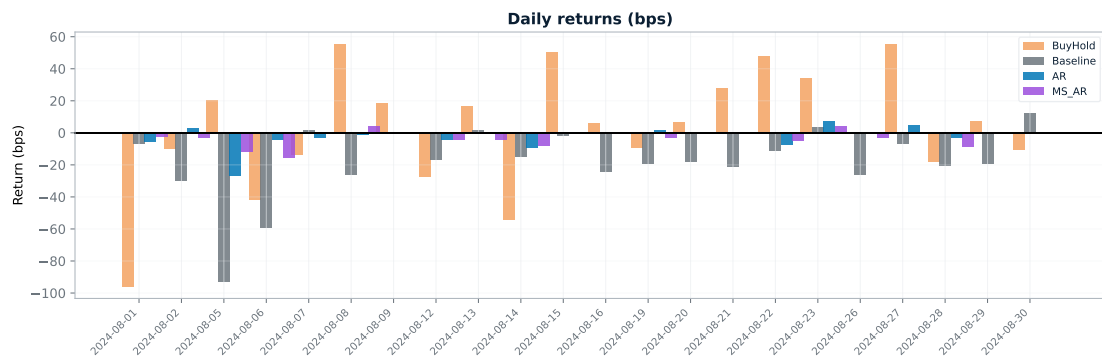


Figure 13: GBP/USD–EUR/USD, August 2024: period returns.

8.7 Spread-return distributions

Per-bar spread-return distributions for the two pairs not shown in the body. Each pair occupies two rows (linear y -axis on top, log y -axis below); columns are August 2024, December 2024, and April 2025.

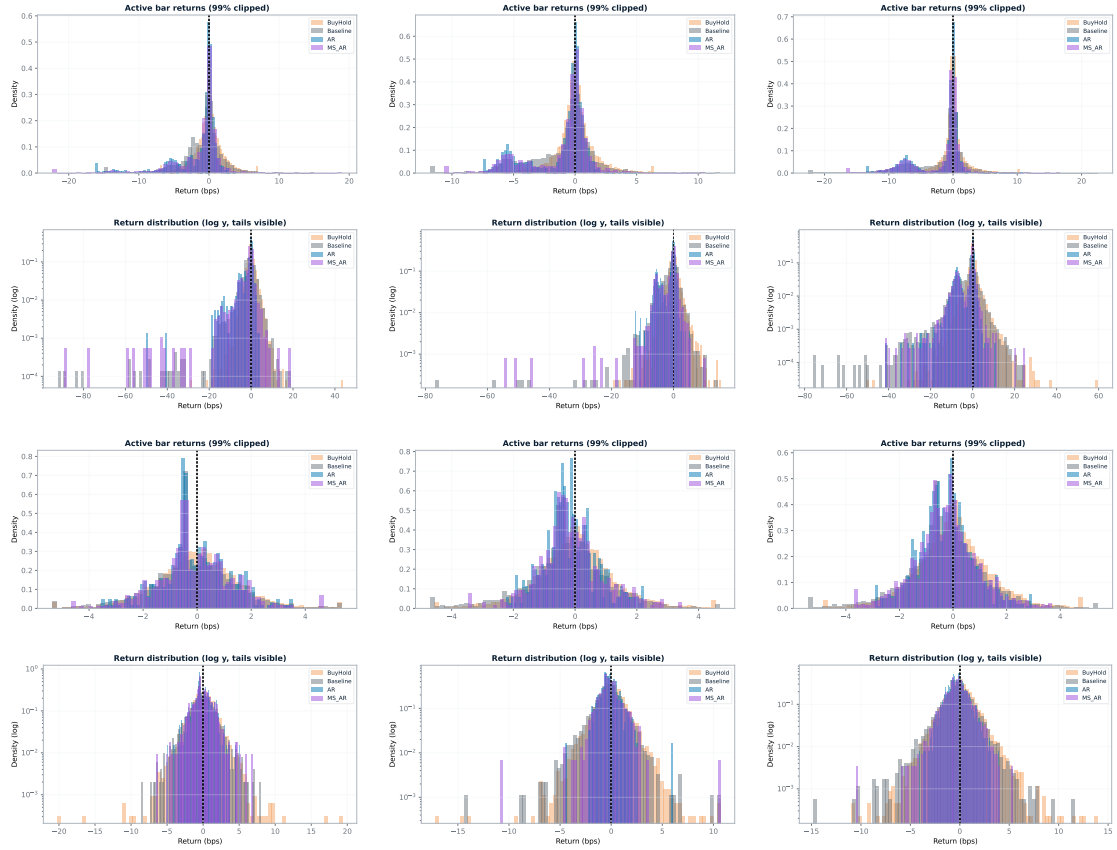


Figure 14: Spread-return distributions for EUR/NOK–EUR/SEK (rows 1–2) and GBP/USD–EUR/USD (rows 3–4). Each pair: top row linear y -axis, bottom row log y -axis. Columns: August 2024 (left), December 2024 (centre), April 2025 (right).

8.8 Markov regime dynamics

Per-day HMM fits showing regime-specific innovation variances $\sigma^{(k)}$, persistence probabilities p_{ii} , and regime means $\mu^{(k)}$, for the August 2024 episode of each pair. The remaining months are available in the companion repository (Furnes [2026](#)).

8.8.1 AUD/USD–NZD/USD

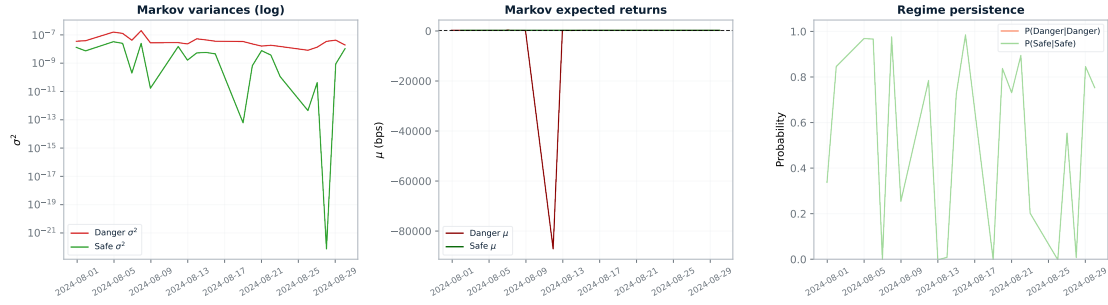


Figure 15: AUD/USD–NZD/USD, August 2024: HMM regime dynamics.

8.8.2 EUR/NOK–EUR/SEK

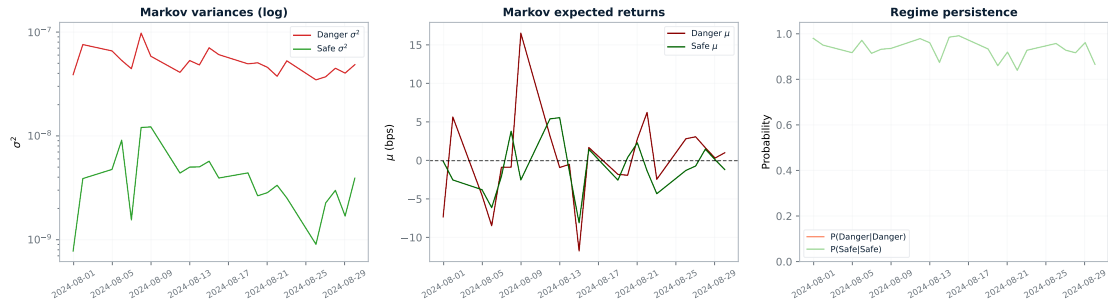


Figure 16: EUR/NOK–EUR/SEK, August 2024: HMM regime dynamics.

8.8.3 GBP/USD–EUR/USD

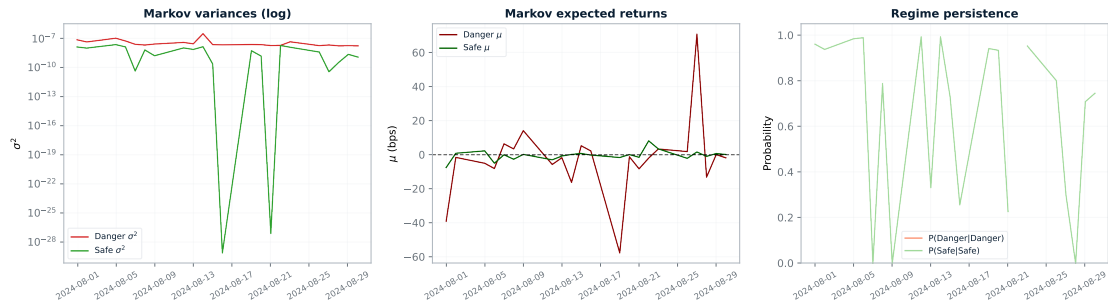


Figure 17: GBP/USD–EUR/USD, August 2024: HMM regime dynamics.

8.9 Daily strategy summaries (raw ledgers)

The three tables that follow are the raw daily ledgers underlying every aggregate number reported in the body and the tearsheets, with one row per trading day, one block per monthly episode, and the same four strategy columns (BH, BL, AR, MS) used throughout the thesis. They are deliberately compact, so the abbreviations are dense; once decoded, every aggregate number in Tables 2–3 and in the tearsheets above can be reconstructed by summing or averaging the corresponding daily entries.

Each cell follows the format **POSITION**, **ACTIVITY**; **PNL**, where **POSITION** describes the position held at the start of the day (L for long, S for short, F/M for flat-or-mixed within the day for the active strategies), **ACTIVITY** reports the trading activity (**NF** for no flips and a flat position the entire day, **nRT** for n completed round-trip trades), and **PNL** is the day’s profit-and-loss in basis points, signed and net of the half-spread plus 0.05 bps commission cost model. The BH column always shows L, **NF** because Buy & Hold simply holds a unit long-spread position throughout, so its daily PnL is the realised ΔS_t aggregated over the day. The BL column shows non-trivial trading activity on every day (the rolling z -score rule flips 5 to 40 times per day depending on volatility), while the AR and MS columns frequently show **NF** because the HMM gate G_t of (16) or the regime-weighted threshold of (18) suppresses entries in the drifting regime. The bold **Total** row at the end of each monthly block is the sum of the corresponding daily PnL column for that strategy in that month, and matches the total-return row of the tearsheets up to floating-point rounding.

Table 10: Daily AUD/USD–NZD/USD strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -31.34	F/M, 11RT; -87.51	F/M, NF; +0.00	F/M, NF; +0.00
08-02	L, NF; +0.48	F/M, 13RT; -61.19	F/M, NF; +0.00	F/M, 1RT; -3.14
08-05	L, NF; +53.17	F/M, 25RT; -213.20	F/M, 16RT; -88.65	F/M, 14RT; -87.77
08-06	L, NF; -39.24	F/M, 19RT; -90.09	F/M, 11RT; -38.79	F/M, 12RT; -61.67
08-07	L, NF; -29.49	F/M, 12RT; -25.74	F/M, 2RT; -0.34	F/M, 2RT; -10.24
08-08	L, NF; +123.49	F/M, 10RT; -78.58	F/M, 7RT; -47.92	F/M, 9RT; -42.99
08-09	L, NF; +4.02	F/M, 4RT; -28.14	F/M, NF; +0.00	F/M, NF; +0.00
08-12	L, NF; -9.05	F/M, 3RT; -4.94	F/M, NF; +0.00	F/M, NF; +0.00
08-13	L, NF; +7.00	F/M, 5RT; -16.18	F/M, NF; +0.00	F/M, NF; +0.00
08-14	L, NF; -19.86	F/M, 8RT; -50.83	F/M, 1RT; -2.11	F/M, 2RT; +7.35
08-15	L, NF; +2.26	F/M, 7RT; -29.29	F/M, NF; +0.00	F/M, NF; +0.00
08-16	L, NF; +21.36	F/M, 3RT; -29.23	F/M, 1RT; -1.48	F/M, 2RT; -6.00
08-19	L, NF; +9.80	F/M, 5RT; -18.18	F/M, NF; +0.00	F/M, NF; +0.00
08-20	L, NF; -5.42	F/M, 6RT; -32.01	F/M, NF; +0.00	F/M, NF; +0.00
08-21	L, NF; -18.57	F/M, 4RT; -26.17	F/M, NF; +0.00	F/M, 1RT; -1.36
08-22	L, NF; -45.36	F/M, 9RT; -14.86	F/M, NF; +0.00	F/M, 1RT; -0.25
08-23	L, NF; -18.31	F/M, 5RT; -24.71	F/M, NF; +0.00	F/M, NF; +0.00
08-26	L, NF; +2.29	F/M, 9RT; -21.43	F/M, NF; +0.00	F/M, NF; +0.00
08-27	L, NF; -25.98	F/M, 6RT; -31.83	F/M, NF; +0.00	F/M, NF; +0.00
08-28	L, NF; -42.90	F/M, 7RT; -24.97	F/M, NF; +0.00	F/M, NF; +0.00
08-29	L, NF; +60.27	F/M, 8RT; -41.63	F/M, NF; +0.00	F/M, NF; +0.00
08-30	L, NF; -44.46	F/M, 5RT; -15.78	F/M, NF; +0.00	F/M, NF; +0.00
Total	-45.80	-966.50	-179.29	-206.08
December 2024				
12-02	L, NF; -23.45	F/M, 13RT; -53.50	F/M, 1RT; -3.17	F/M, 1RT; -0.21
12-03	L, NF; +14.45	F/M, 6RT; -8.46	F/M, 1RT; -1.22	F/M, 1RT; -1.22
12-04	L, NF; +10.09	F/M, 12RT; -55.87	F/M, NF; +0.00	F/M, 1RT; -1.64
12-05	L, NF; -13.59	F/M, 7RT; -23.94	F/M, NF; +0.00	F/M, NF; +0.00
12-06	L, NF; -6.79	F/M, 10RT; -41.15	F/M, NF; +0.00	F/M, NF; +0.00
12-09	L, NF; +22.70	F/M, 6RT; -21.88	F/M, NF; +0.00	F/M, NF; +0.00
12-10	L, NF; -11.36	F/M, 6RT; -6.40	F/M, NF; +0.00	F/M, 1RT; +0.32
12-11	L, NF; +7.31	F/M, 17RT; -63.10	F/M, 5RT; -13.85	F/M, 6RT; -26.97
12-12	L, NF; +2.42	F/M, 12RT; -45.77	F/M, 5RT; -17.46	F/M, 5RT; -15.63
12-13	L, NF; +5.05	F/M, 8RT; -29.26	F/M, NF; +0.00	F/M, NF; +0.00
12-16	L, NF; -22.37	F/M, 7RT; -15.01	F/M, NF; +0.00	F/M, NF; +0.00
12-17	L, NF; -8.15	F/M, 5RT; -17.33	F/M, 1RT; -3.57	F/M, 1RT; -1.71
12-18	L, NF; -6.43	F/M, 6RT; -73.73	F/M, 5RT; -7.48	F/M, 6RT; -29.82
12-19	L, NF; +16.30	F/M, 12RT; -68.77	F/M, 7RT; -48.28	F/M, 6RT; -23.91
12-20	L, NF; -23.04	F/M, 11RT; -42.48	F/M, 6RT; -17.80	F/M, 5RT; -20.30
12-23	L, NF; -9.43	F/M, 8RT; -28.51	F/M, 1RT; -2.02	F/M, 1RT; -2.02
12-24	L, NF; -5.02	F/M, 4RT; -30.63	F/M, NF; +0.00	F/M, NF; +0.00
12-26	L, NF; +12.32	F/M, 7RT; -45.35	F/M, 4RT; -16.94	F/M, 5RT; -33.32
12-27	L, NF; -16.39	F/M, 7RT; -36.77	F/M, NF; +0.00	F/M, NF; +0.00
12-30	L, NF; -14.99	F/M, 6RT; -42.23	F/M, 1RT; -4.92	F/M, 1RT; -1.59
12-31	L, NF; +31.04	F/M, 9RT; -26.34	F/M, NF; +0.00	F/M, NF; +0.00
Total	-39.30	-776.48	-136.71	-158.00
April 2025				
04-01	L, NF; +0.13	F/M, 6RT; -38.05	F/M, NF; +0.00	F/M, NF; +0.00
04-02	L, NF; -21.94	F/M, 14RT; -60.07	F/M, NF; +0.00	F/M, 1RT; +0.00
04-03	L, NF; -6.77	F/M, 17RT; -91.15	F/M, NF; +0.00	F/M, 1RT; -5.26
04-04	L, NF; -86.88	F/M, 34RT; -165.76	F/M, 17RT; -93.99	F/M, 15RT; -81.47
04-07	L, NF; +52.62	F/M, 44RT; -328.86	F/M, 25RT; -155.86	F/M, 28RT; -192.20
04-08	L, NF; +19.78	F/M, 26RT; -116.89	F/M, 18RT; -77.60	F/M, 17RT; -74.08
04-09	L, NF; +52.03	F/M, 38RT; -166.21	F/M, 18RT; -74.35	F/M, 19RT; -85.66
04-10	L, NF; -68.00	F/M, 29RT; -227.84	F/M, 8RT; -36.89	F/M, 11RT; -51.01
04-11	L, NF; +52.30	F/M, 38RT; -203.31	F/M, 18RT; -75.91	F/M, 14RT; -51.31
04-14	L, NF; +17.40	F/M, 18RT; -82.49	F/M, 6RT; -29.42	F/M, 7RT; -27.67
04-15	L, NF; -6.36	F/M, 12RT; -66.81	F/M, NF; +0.00	F/M, 2RT; -18.00
04-16	L, NF; +20.97	F/M, 16RT; -63.10	F/M, 14RT; -43.28	F/M, 14RT; -57.86
04-17	L, NF; -5.12	F/M, 9RT; -83.37	F/M, 2RT; -6.23	F/M, 3RT; -13.96
04-18	L, NF; +55.04	F/M, 48RT; -403.09	F/M, 7RT; -52.00	F/M, 10RT; -75.44
04-21	L, NF; +2.23	F/M, 10RT; -40.86	F/M, 1RT; -2.54	F/M, NF; +5.08
04-22	L, NF; -7.56	F/M, 14RT; -77.44	F/M, 1RT; +1.58	F/M, NF; +0.00
04-23	L, NF; +31.92	F/M, 13RT; -48.66	F/M, NF; +0.00	F/M, NF; +0.00
04-24	L, NF; +4.48	F/M, 13RT; -50.70	F/M, NF; +0.00	F/M, NF; +0.00
04-25	L, NF; +38.03	F/M, 9RT; -71.88	F/M, NF; +0.00	F/M, NF; +0.00
04-28	L, NF; +41.98	F/M, 7RT; -52.53	F/M, 1RT; -2.49	F/M, 1RT; -2.49
04-29	L, NF; -26.78	F/M, 6RT; -33.65	F/M, 1RT; -2.04	F/M, 1RT; +1.13
04-30	L, NF; -4.29	F/M, 7RT; -48.66	F/M, 1RT; -1.21	F/M, 1RT; -1.13
Total	+155.20	-2521.40	-652.20	-731.30

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

Table 11: Daily EUR/NOK–EUR/SEK strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -18.20	F/M, 29RT; -446.12	F/M, 7RT; -201.46	F/M, 8RT; -284.68
08-02	L, NF; +145.51	F/M, 36RT; -293.29	F/M, 12RT; -150.88	F/M, 13RT; -145.72
08-05	L, NF; -59.15	F/M, 65RT; -839.32	F/M, 13RT; -159.51	F/M, 16RT; -218.97
08-06	L, NF; -129.52	F/M, 45RT; -546.57	F/M, 10RT; -168.80	F/M, 11RT; -163.92
08-07	L, NF; -0.51	F/M, 37RT; -466.82	F/M, 6RT; -112.94	F/M, 8RT; -235.61
08-08	L, NF; -16.30	F/M, 24RT; -513.76	F/M, 7RT; -63.19	F/M, 9RT; -253.59
08-09	L, NF; -40.90	F/M, 21RT; -143.92	F/M, 7RT; -36.39	F/M, 7RT; -44.56
08-12	L, NF; -24.75	F/M, 21RT; -238.40	F/M, 7RT; -81.19	F/M, 8RT; -138.30
08-13	L, NF; -3.94	F/M, 23RT; -251.14	F/M, 10RT; -124.40	F/M, 11RT; -195.03
08-14	L, NF; +4.43	F/M, 12RT; -82.69	F/M, 1RT; -7.49	F/M, 2RT; -27.24
08-15	L, NF; -43.86	F/M, 24RT; -200.67	F/M, 11RT; -130.40	F/M, 10RT; -73.57
08-16	L, NF; +34.20	F/M, 15RT; -22.71	F/M, 1RT; -7.87	F/M, 1RT; -7.87
08-19	L, NF; -26.30	F/M, 27RT; -243.01	F/M, 8RT; -115.31	F/M, 12RT; -120.18
08-20	L, NF; -43.63	F/M, 27RT; -218.99	F/M, NF; +0.00	F/M, 3RT; -12.75
08-21	L, NF; +3.32	F/M, 27RT; -276.27	F/M, 5RT; -96.67	F/M, 5RT; -94.16
08-22	L, NF; +50.77	F/M, 12RT; -97.74	F/M, NF; +0.00	F/M, 1RT; -4.31
08-23	L, NF; -25.10	F/M, 18RT; -157.05	F/M, 4RT; -36.26	F/M, 3RT; -24.21
08-26	L, NF; +51.45	F/M, 32RT; -277.93	F/M, 8RT; -88.90	F/M, 9RT; -92.07
08-27	L, NF; +0.63	F/M, 27RT; -253.49	F/M, 4RT; -63.65	F/M, 5RT; -69.94
08-28	L, NF; -1.52	F/M, 25RT; -279.21	F/M, 4RT; -37.30	F/M, 4RT; -40.84
08-29	L, NF; +2.24	F/M, 26RT; -302.57	F/M, 8RT; -111.44	F/M, 11RT; -222.58
08-30	L, NF; -7.39	F/M, 27RT; -125.76	F/M, 3RT; -22.69	F/M, 5RT; -20.42
Total	-148.52	-6277.43	-1816.74	-2490.52
December 2024				
12-02	L, NF; -39.79	F/M, 29RT; -313.72	F/M, 6RT; -61.49	F/M, 5RT; -68.03
12-03	L, NF; -11.34	F/M, 27RT; -271.73	F/M, 17RT; -163.55	F/M, 19RT; -218.26
12-04	L, NF; +26.20	F/M, 27RT; -256.08	F/M, 9RT; -88.75	F/M, 12RT; -118.08
12-05	L, NF; +94.66	F/M, 27RT; -275.13	F/M, 15RT; -171.88	F/M, 18RT; -237.77
12-06	L, NF; +34.66	F/M, 25RT; -164.41	F/M, 6RT; -66.95	F/M, 6RT; -63.32
12-09	L, NF; -60.71	F/M, 21RT; -196.73	F/M, 7RT; -64.80	F/M, 7RT; -59.77
12-10	L, NF; +7.11	F/M, 22RT; -224.43	F/M, 12RT; -140.26	F/M, 12RT; -145.91
12-11	L, NF; +8.96	F/M, 17RT; -185.45	F/M, 6RT; -52.74	F/M, 9RT; -102.84
12-12	L, NF; -22.14	F/M, 22RT; -220.72	F/M, 8RT; -78.50	F/M, 8RT; -77.38
12-13	L, NF; +20.32	F/M, 21RT; -174.68	F/M, 6RT; -51.65	F/M, 7RT; -57.23
12-16	L, NF; +12.13	F/M, 26RT; -266.25	F/M, 14RT; -131.47	F/M, 15RT; -135.04
12-17	L, NF; +11.70	F/M, 15RT; -182.22	F/M, 6RT; -46.96	F/M, 9RT; -69.45
12-18	L, NF; +19.65	F/M, 17RT; -182.89	F/M, 6RT; -62.83	F/M, 6RT; -95.34
12-19	L, NF; +85.01	F/M, 20RT; -163.33	F/M, 8RT; -95.65	F/M, 10RT; -113.75
12-20	L, NF; -122.08	F/M, 26RT; -308.33	F/M, 11RT; -123.09	F/M, 12RT; -152.12
12-23	L, NF; +2.03	F/M, 25RT; -225.71	F/M, 8RT; -97.23	F/M, 8RT; -97.95
12-24	L, NF; +14.16	F/M, 8RT; -248.71	F/M, 1RT; -9.50	F/M, 1RT; -9.50
12-26	L, NF; +95.74	F/M, 14RT; -168.21	F/M, NF; +0.00	F/M, 1RT; -33.15
12-27	L, NF; -54.01	F/M, 16RT; -158.75	F/M, 2RT; -13.16	F/M, 3RT; -28.17
12-30	L, NF; -41.37	F/M, 16RT; -202.59	F/M, 8RT; -65.22	F/M, 10RT; -175.01
12-31	L, NF; -20.30	F/M, 9RT; -83.76	F/M, 1RT; -7.18	F/M, 2RT; -12.69
Total	+60.59	-4473.83	-1592.86	-2070.76
April 2025				
04-01	L, NF; +1.87	F/M, 18RT; -240.30	F/M, 5RT; -49.26	F/M, 5RT; -49.14
04-02	L, NF; +88.21	F/M, 17RT; -241.28	F/M, 9RT; -79.96	F/M, 12RT; -152.18
04-03	L, NF; +97.48	F/M, 37RT; -495.85	F/M, 15RT; -182.36	F/M, 17RT; -214.19
04-04	L, NF; +180.78	F/M, 29RT; -446.44	F/M, 15RT; -171.06	F/M, 17RT; -256.78
04-07	L, NF; +231.98	F/M, 44RT; -1213.70	F/M, 16RT; -410.56	F/M, 15RT; -404.87
04-08	L, NF; +21.78	F/M, 28RT; -454.75	F/M, 11RT; -188.16	F/M, 12RT; -233.40
04-09	L, NF; -56.74	F/M, 50RT; -1071.52	F/M, 12RT; -230.54	F/M, 19RT; -340.36
04-10	L, NF; +361.99	F/M, 58RT; -1203.87	F/M, 23RT; -384.66	F/M, 20RT; -378.45
04-11	L, NF; -190.56	F/M, 53RT; -1108.65	F/M, 21RT; -486.68	F/M, 21RT; -503.42
04-14	L, NF; -119.72	F/M, 57RT; -1186.25	F/M, 24RT; -399.04	F/M, 28RT; -526.91
04-15	L, NF; +49.63	F/M, 64RT; -826.57	F/M, 31RT; -408.54	F/M, 28RT; -373.06
04-16	L, NF; +44.51	F/M, 56RT; -930.52	F/M, 23RT; -449.51	F/M, 30RT; -543.34
04-17	L, NF; -82.92	F/M, 62RT; -836.64	F/M, 38RT; -542.80	F/M, 37RT; -569.53
04-18	L, NF; +33.90	F/M, 84RT; -1957.30	F/M, 27RT; -560.49	F/M, 37RT; -815.25
04-21	L, NF; +1.90	F/M, 71RT; -1346.75	F/M, 43RT; -689.11	F/M, 49RT; -922.87
04-22	L, NF; -87.83	F/M, 62RT; -1181.46	F/M, 24RT; -383.63	F/M, 29RT; -472.88
04-23	L, NF; +45.25	F/M, 83RT; -1573.22	F/M, 33RT; -493.71	F/M, 36RT; -622.10
04-24	L, NF; -21.95	F/M, 54RT; -780.65	F/M, 28RT; -446.71	F/M, 29RT; -439.88
04-25	L, NF; -21.48	F/M, 42RT; -591.91	F/M, 27RT; -410.07	F/M, 26RT; -369.03
04-28	L, NF; -65.16	F/M, 35RT; -565.56	F/M, 17RT; -256.15	F/M, 19RT; -334.78
04-29	L, NF; +13.06	F/M, 51RT; -913.39	F/M, 27RT; -497.57	F/M, 27RT; -489.51
04-30	L, NF; +1.70	F/M, 54RT; -739.84	F/M, 25RT; -382.90	F/M, 30RT; -475.34
Total	+527.68	-19906.42	-8103.47	-9487.27

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

Table 12: Daily GBP/USD–EUR/USD strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -95.81	F/M, 16RT; -7.01	F/M, 7RT; -5.37	F/M, 7RT; -2.36
08-02	L, NF; -9.79	F/M, 18RT; -29.83	F/M, 5RT; +2.74	F/M, 5RT; -3.24
08-05	L, NF; +20.45	F/M, 38RT; -93.04	F/M, 24RT; -26.81	F/M, 23RT; -11.86
08-06	L, NF; -41.77	F/M, 20RT; -59.23	F/M, 8RT; -4.02	F/M, 9RT; -15.54
08-07	L, NF; -13.88	F/M, 14RT; +1.63	F/M, 2RT; -3.30	F/M, 3RT; +0.35
08-08	L, NF; +55.12	F/M, 13RT; -25.94	F/M, 1RT; -1.33	F/M, 2RT; +4.34
08-09	L, NF; +18.42	F/M, 6RT; -0.35	F/M, NF; +0.00	F/M, NF; +0.00
08-12	L, NF; -27.43	F/M, 5RT; -16.67	F/M, 2RT; -4.26	F/M, 2RT; -4.26
08-13	L, NF; +16.26	F/M, 11RT; +1.41	F/M, NF; +0.00	F/M, 1RT; -4.32
08-14	L, NF; -54.31	F/M, 11RT; -14.84	F/M, 8RT; -8.98	F/M, 9RT; -8.11
08-15	L, NF; +50.15	F/M, 7RT; -1.52	F/M, NF; +0.00	F/M, 1RT; +0.18
08-16	L, NF; +5.67	F/M, 5RT; -24.36	F/M, NF; +0.00	F/M, NF; +0.00
08-19	L, NF; -8.97	F/M, 8RT; -19.09	F/M, 2RT; +1.47	F/M, 2RT; -3.12
08-20	L, NF; +6.62	F/M, 11RT; -18.05	F/M, NF; +0.00	F/M, NF; +0.00
08-21	L, NF; +27.71	F/M, 13RT; -21.26	F/M, NF; +0.00	F/M, NF; +0.00
08-22	L, NF; +47.55	F/M, 14RT; -11.02	F/M, 13RT; -7.55	F/M, 13RT; -4.86
08-23	L, NF; +34.32	F/M, 12RT; +3.20	F/M, 4RT; +7.36	F/M, 3RT; +4.26
08-26	L, NF; +0.52	F/M, 8RT; -25.83	F/M, NF; +0.00	F/M, 1RT; -2.85
08-27	L, NF; +55.37	F/M, 11RT; -6.69	F/M, 1RT; +4.48	F/M, NF; +0.00
08-28	L, NF; -18.20	F/M, 12RT; -20.50	F/M, 1RT; -3.03	F/M, 1RT; -8.76
08-29	L, NF; +7.45	F/M, 12RT; -19.10	F/M, NF; +0.00	F/M, NF; +0.00
08-30	L, NF; -10.38	F/M, 13RT; +12.00	F/M, NF; +0.00	F/M, NF; +0.00
Total	+65.07	-396.09	-48.60	-60.15
December 2024				
12-02	L, NF; +3.19	F/M, 27RT; -58.38	F/M, 8RT; -8.08	F/M, 8RT; -18.27
12-03	L, NF; -12.52	F/M, 18RT; -40.69	F/M, 6RT; -16.33	F/M, 6RT; -16.33
12-04	L, NF; +15.83	F/M, 21RT; -61.39	F/M, NF; +0.00	F/M, NF; +0.00
12-05	L, NF; -3.05	F/M, 16RT; -2.69	F/M, 1RT; -0.72	F/M, 1RT; -0.72
12-06	L, NF; +13.69	F/M, 17RT; -26.09	F/M, NF; +0.00	F/M, 1RT; +0.53
12-09	L, NF; -5.84	F/M, 15RT; -37.29	F/M, NF; +0.00	F/M, NF; +0.00
12-10	L, NF; +60.00	F/M, 15RT; -25.24	F/M, NF; +0.00	F/M, NF; +0.00
12-11	L, NF; -1.06	F/M, 18RT; -23.30	F/M, 2RT; +1.53	F/M, 3RT; +2.34
12-12	L, NF; -20.80	F/M, 19RT; -67.59	F/M, 7RT; -10.71	F/M, 8RT; -17.29
12-13	L, NF; -73.44	F/M, 11RT; -24.92	F/M, 9RT; -16.17	F/M, 9RT; -10.90
12-16	L, NF; +31.93	F/M, 13RT; -18.32	F/M, NF; +0.00	F/M, 2RT; +1.07
12-17	L, NF; +46.60	F/M, 12RT; -5.50	F/M, 6RT; +5.57	F/M, 5RT; -1.47
12-18	L, NF; +5.93	F/M, 16RT; -45.21	F/M, 7RT; -7.77	F/M, 7RT; -6.00
12-19	L, NF; -68.70	F/M, 21RT; -48.72	F/M, 2RT; -10.51	F/M, 3RT; -7.08
12-20	L, NF; +18.33	F/M, 17RT; -39.41	F/M, 1RT; -4.96	F/M, 1RT; -2.41
12-23	L, NF; +10.12	F/M, 12RT; -38.83	F/M, 4RT; -2.36	F/M, 5RT; -0.32
12-24	L, NF; +20.19	F/M, 5RT; -36.10	F/M, NF; +0.00	F/M, NF; +0.00
12-26	L, NF; -33.41	F/M, 8RT; -38.63	F/M, 1RT; +0.14	F/M, 1RT; +2.41
12-27	L, NF; +32.86	F/M, 11RT; -53.24	F/M, NF; +0.00	F/M, NF; -2.11
12-30	L, NF; -29.32	F/M, 12RT; -28.81	F/M, 1RT; +0.73	F/M, 1RT; +0.73
12-31	L, NF; +14.58	F/M, 11RT; -10.05	F/M, NF; +0.00	F/M, NF; +0.00
Total	+25.11	-730.40	-69.64	-75.82
April 2025				
04-01	L, NF; -10.53	F/M, 13RT; -33.50	F/M, NF; +0.00	F/M, NF; +0.00
04-02	L, NF; +42.65	F/M, 19RT; -63.97	F/M, 1RT; +1.38	F/M, 1RT; -9.69
04-03	L, NF; -35.01	F/M, 31RT; -40.97	F/M, NF; +0.00	F/M, 1RT; -0.94
04-04	L, NF; -126.74	F/M, 36RT; -108.78	F/M, 9RT; -17.31	F/M, 13RT; -40.84
04-07	L, NF; -98.79	F/M, 49RT; -99.90	F/M, 27RT; -74.31	F/M, 27RT; -69.84
04-08	L, NF; +30.71	F/M, 33RT; -67.22	F/M, NF; +0.00	F/M, 2RT; -7.76
04-09	L, NF; +43.84	F/M, 48RT; -196.25	F/M, 16RT; -35.22	F/M, 16RT; -33.64
04-10	L, NF; +78.06	F/M, 36RT; -145.58	F/M, 16RT; -50.82	F/M, 16RT; -61.12
04-11	L, NF; +57.23	F/M, 49RT; -151.21	F/M, 17RT; -16.63	F/M, 16RT; -7.59
04-14	L, NF; +35.17	F/M, 22RT; -44.74	F/M, 11RT; -4.81	F/M, 13RT; -21.99
04-15	L, NF; +54.20	F/M, 19RT; -23.08	F/M, NF; +0.00	F/M, 1RT; +0.32
04-16	L, NF; -75.64	F/M, 26RT; -45.62	F/M, NF; +0.00	F/M, 1RT; -2.60
04-17	L, NF; +22.71	F/M, 19RT; -25.93	F/M, 9RT; -20.24	F/M, 5RT; -14.00
04-18	L, NF; -4.82	F/M, 52RT; -183.09	F/M, 27RT; -85.29	F/M, 27RT; -84.64
04-21	L, NF; -0.61	F/M, 20RT; -31.85	F/M, 1RT; -2.77	F/M, 1RT; +1.90
04-22	L, NF; -32.08	F/M, 25RT; -71.98	F/M, 9RT; -13.07	F/M, 8RT; -6.13
04-23	L, NF; -14.82	F/M, 22RT; -45.89	F/M, 5RT; -15.40	F/M, 5RT; -10.53
04-24	L, NF; +46.08	F/M, 15RT; -6.51	F/M, NF; +0.00	F/M, NF; +0.00
04-25	L, NF; -17.01	F/M, 13RT; -23.75	F/M, 3RT; -16.41	F/M, 2RT; -16.47
04-28	L, NF; +68.55	F/M, 15RT; -48.12	F/M, NF; +0.00	F/M, NF; +0.00
04-29	L, NF; -2.26	F/M, 10RT; -27.93	F/M, 1RT; -1.85	F/M, 1RT; -6.11
04-30	L, NF; -44.94	F/M, 10RT; -23.28	F/M, 1RT; +2.14	F/M, 4RT; -14.26
Total	+15.95	-1509.15	-350.61	-405.93

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

8.10 Symbol dictionary

The table below collects the mathematical notation used in the methods sections, in the order in which the symbols are introduced. Where two symbols refer to the same quantity in slightly different parameterisations (for example $c^{(k)}$ versus $\mu^{(k)}$ for the MS-AR(1) intercept), both rows are kept so the dictionary can be used as a lookup reference for either form.

Symbol	Meaning
<i>Data and pre-processing (Section 2).</i>	
$i \in \{A, B\}$	Index of the two legs of a currency pair.
t	Time index of a pre-averaged bar (block of L ticks).
$P_t^{\text{bid}, i}$	Bid price of leg i at time t .
$P_t^{\text{ask}, i}$	Ask price of leg i at time t .
$P_t^{\text{mid}, i}$	Mid-price, $\frac{1}{2}(P_t^{\text{bid}} + P_t^{\text{ask}})$, see (1).
HS_t^i	Basis-point half-spread of leg i at time t , see (1).
L	Pre-averaging block size in number of ticks, see (2).
$\bar{P}_t^i$	Pre-averaged mid-price of leg i at block index t , see (2).
r_t^i	Log return of leg i at bar t , $\log P_t^{\text{mid}, i} - \log P_{t-1}^{\text{mid}, i}$.
<i>Baseline pairs-trading rule (Section 3).</i>	
β	Cointegration hedge ratio (global), see (3).
β_t, α_t	Rolling hedge ratio and intercept on window $\mathcal{W}_t$, see (4).
W_β	Length of the rolling OLS window for (β_t, α_t) .
$\mathcal{W}_t$	Index set $\{t - W_\beta + 1, \dots, t\}$ used by the rolling estimator.
$\bar{x}_t, \bar{y}_t$	Window means of $x_s = \log P_s^{\text{mid}, B}$ and $y_s = \log P_s^{\text{mid}, A}$.
S_t	Cointegration spread $\log P_t^{\text{mid}, A} - \beta_t \log P_t^{\text{mid}, B} - \alpha_t$.
ΔS_t	Spread return $r_t^A - \beta_{t-1} r_t^B$, see (5).
W_z	Length of the rolling window for the z -score mean and variance.
μ_t^S, σ_t^S	Rolling mean and standard deviation of S_t over the W_z -bar window.
Z_t	Rolling z -score $(S_t - \mu_t^S) / \sigma_t^S$, see (6).
z_q	Entry threshold on $ Z_t $ for the baseline rule.
z_x	Exit threshold on $ Z_t $ for the baseline rule.
π_t	$\in$ Position at bar t (short / flat / long the spread), see (7).
$\{-1, 0, +1\}$	
π_t^{BH}	Buy-and-Hold reference position, $\pi_t^{\text{BH}} \equiv +1$.
π_t^{baseline}	Baseline-rule position produced by (7).
<i>Regime-aware models (Section 4).</i>	
p	Order of the generic AR(p) model in (8).
c, ϕ_j, σ^2	Intercept, AR coefficients, and innovation variance of AR(p); for $p = 1$, $\phi_1 \equiv \phi$, see (9).
ε_t	Innovation in the AR(p)/AR(1) model, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.
K	Number of regimes in the Markov chain.

Symbol	Meaning
K_t	$\in$ Hidden regime at bar t .
$\{1, \dots, K\}$	
$\mathbf{P} = (p_{ij})$	$K \times K$ transition matrix, $p_{ij} = \mathbb{P}(K_t = j \mid K_{t-1} = i)$, see (10).
ν	$=$ Initial-state distribution of the Markov chain.
$(\nu_1, \dots, \nu_K)$	
$c^{(k)}, \rho^{(k)}, \sigma^{(k)}$	Regime- k intercept, AR(1) coefficient, and innovation standard deviation in (11).
$\mu^{(k)}$	Regime- k unconditional mean, $\mu^{(k)} = c^{(k)} / (1 - \rho^{(k)})$, see (12).
Θ	Full HMM parameter set $\{\nu, \mathbf{P}, c^{(k)}, \rho^{(k)}, \sigma^{(k)}\}_{k=1}^K$, see (13).
$\gamma_t(k)$	Smoothed regime posterior $\mathbb{P}(K_t = k \mid S_1, \dots, S_T; \Theta)$, see (14).
$k_{\text{MR}}, k_{\text{DR}}$	Indices of the mean-reverting and drifting regimes, by innovation variance, see (15).
$\gamma_t^{\text{MR}}, \gamma_t^{\text{DR}}$	Posterior probabilities of the MR and DR regimes at bar t ; $\gamma_t^{\text{MR}} + \gamma_t^{\text{DR}} = 1$.
$\delta \in (0, 1)$	Danger-threshold tolerance in the AR gate, see (16).
$G_t \in \{0, 1\}$	Binary AR gate, $G_t = \mathbb{1}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}$.
z_v	MS-AR drifting-regime threshold, with $z_v \geq z_q$, see (18).
$z_q^{\text{eff}}(t)$	Regime-weighted effective entry threshold for MS-AR, see (18).
<i>Back-test protocol and performance metrics (Section 5).</i>	
R_t	Strategy return at bar t .
μ_R, σ_R	Mean and standard deviation of R_t .
R_f	Benchmark risk-free rate, set to zero throughout.
T	Number of bars per year used in the annualisation of σ_R .
σ_{ann}	Annualised return volatility, $\sigma_R \sqrt{T}$, see (19).
$\mathcal{S}, \mathcal{K}$	Skewness and kurtosis of R_t , see (20).
α	Confidence level for VaR/CVaR, $\alpha = 0.95$, see (21).
VaR_α	Value-at-Risk at level α , $-Q_{1-\alpha}(R)$.
CVaR_α	Conditional VaR (expected shortfall) at level α .
TF	Tail-fatness ratio $\text{CVaR}_\alpha / \text{VaR}_\alpha$, see (21).
V_t, V_t^*	Portfolio value at bar t and its running peak $\max_{s \leq t} V_s$.
D_t	Drawdown $(V_t^* - V_t) / V_t^*$, see (22).
MDD	Maximum drawdown of D_t , see (19).
UI	Ulcer index of Martin and McCann (1989), see (22).
SR	Sharpe ratio $(\mu_R - R_f) / \sigma_R$, see (19).
Sortino	Sortino ratio $(\mu_R - R_f) / \sigma_d$, with σ_d the downside deviation.
Calmar	Calmar ratio, annualised return divided by maximum drawdown.
AR	Annualised return used in the Calmar ratio. (Not to be confused with the AR-gated strategy.)